

## Esmuflily - SMuFL in LilyPond

Esmuflily is an extension for [LilyPond](#) that supports [SMuFL](#) compliant fonts like [Bravura](#) or [Ekmelos](#). It provides [switches](#) to turn on the SMuFL support for individual types of graphical objects (clefs, note heads, flags, etc.) and it defines additional commands and styles for SMuFL glyphs. So scores can benefit from both, SMuFL's comprehensive character set and LilyPond's awesome Emmentaler font.

Accidentals and key signatures are omitted. They are supported in detail by [Ekmelily](#). See [Cosmuflily](#) for how to combine both.

Esmuflily requires LilyPond version 2.24 or higher.

This document uses the [Ekmelos](#) font for all SMuFL glyph.

16 September 2026

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## Author and License

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## Usage

---

### [Esmuflily on GitHub](#)

Install from folder [ly](#) the following files:

- `esmufl.ily`
- `ekmd.scm`
- `ekmd-template.scm`

`ekmd-template.scm` is required only if no [metadata cache file](#) exists for the desired [font](#).

For some fonts, a cache file is already available:

- `ekmd-bravura.scm`
- `ekmd-ekmelos.scm`
- `ekmd-leland.scm`
- `ekmd-petaluma.scm`
- `ekmd-sebastian.scm`

Add the following lines near the top of your LilyPond input file.

`ekmFont` is optional but must be placed before `\include`.

```
ekmFont = "FONT"  
\include "esmufl.ily"
```

### Esmuflily + Ekmelily

See [Cosmuflily](#) to combine Esmuflily with [Ekmelily](#) for a full SMuFL support.

## Fonts

---

Esmuflily requires a [SMuFL](#) compliant font.

It uses [Ekmelos](#) by default.

Another font can be selected, either with the variable preceding the include file

```
ekmFont = "FONT"
```

or with the command line option

```
-dekmfont=FONT
```

Note that both, Esmuflily and [Ekmelily](#) select a font that way. Separate fonts for graphical objects can be specified with the `font-name` property, such as

```
\override NoteHead.font-name = #"FONT"
```

### Drawing paths

Esmuflily supports drawing paths instead of font glyphs, which allows to produce stand-alone SVG output. A trailing # in FONT switches to globally drawing paths, i.e. it effects all SMuFL output. It includes the file `FNAME-paths.ily` if available, where FNAME is the font name in all lowercase. This file must provide the Scheme procedure `ekm-path-stencil`.

Such a file can be generated with `pathable.py`. See `ekmelos-paths.ily` for the [Ekmelos](#) font.

Note that paths do not support:

- Spaces and other glyphs without a contour.
- Font features like stylistic alternates and ligatures.
- Side-bearing.

Example:

```
\ekm-chars #'(#xE262 #xE566 #xEAA6 #xEAA5)
```



The path (SVG) output ignores the ligature of sharp (U+E262) and trill (U+E566), and the side-bearing of the trill span segments, so they do not overlap.

Note: The [markup commands](#) `\ekm-charf` and `\ekm-str` can draw paths independent of this global setting, but still require `ekm-path-stencil`.

The markup command `\ekm-lily` cannot draw paths.

## Font Metadata

---

Esmuflily can use font-specific metadata provided by a [SMuFL](#) compliant [JSON](#) file or by a cache file. If the latter doesn't exist, the JSON file is read and a cache file is created to be used subsequently.

Note: In the JSON file, glyphs must be given with their canonical glyph name, not with the Unicode code point.

A cache file is a Scheme file with the metadata of interest for Esmuflily extracted from a JSON file, and additional data for font-specific [musical symbols](#).

### JSON file locations

See also [SMuFL](#) "Font-specific metadata locations" .

- MD\_LOCATION/MD\_NAME
- MD\_LOCATION
  1. PRIVATE\_LOCATION
  2. USER\_LOCATION/SMuFL/Fonts/FONT
  3. SYSTEM\_LOCATION/SMuFL/Fonts/FONT
- PRIVATE\_LOCATION  
Can be specified either with the variable preceding the include file

```
ekmMetadata = "PRIVATE_LOCATION"
```

or with the command line option

```
-dekmmetadata=PRIVATE_LOCATION
```

A trailing # in PRIVATE\_LOCATION forces creating a cache file even if it already exists.

- USER\_LOCATION
 

|         |                               |
|---------|-------------------------------|
| Linux   | \$XDG_DATA_HOME               |
| Windows | %LOCALAPPDATA%                |
| macOS   | ~/Library/Application Support |
- SYSTEM\_LOCATION
 

|         |                              |
|---------|------------------------------|
| Linux   | \$XDG_DATA_DIRS              |
| Windows | %CommonProgramFiles%         |
|         | %CommonProgramFiles(x86)%    |
| macOS   | /Library/Application Support |
- MD\_NAME
  1. FNAME\_metadata.json
  2. FNAME.json
  3. metadata.json
- FONT  
Font name in normal spelling, eg. Bravura
- FNAME  
Font name in all lowercase, eg. bravura

## Cache file locations

- EKMD\_LOCATION/EKMD\_NAME
- EKMD\_LOCATION
  1. LilyPond include directory, usually the location of `esmuf1.ily`
  2. PRIVATE\_LOCATION
  3. USER\_LOCATION/SMuFL/Fonts/FONT
  4. SYSTEM\_LOCATION/SMuFL/Fonts/FONT

- EKMD\_NAME

|                                 |                                    |
|---------------------------------|------------------------------------|
| <code>ekmd-FNAME.scn</code>     | Cache file for the font FNAME      |
| <code>ekmd-template.scn</code>  | Template cache file for new fonts  |
| <code>types-FNAME.scn</code>    | Types table for the font FNAME     |
| <code>types-template.scn</code> | Template types table for new fonts |

## Commands

---

Some commands with a corresponding LilyPond command are simpler implemented, e.g. they ignore properties, while some other commands provide additional features.

Most of the commands, in particular, all markup commands always produce SMuFL output, independent of any [SMuFL switches](#). Other commands behave differently when the corresponding switch is turned off:

[\[LP\]](#) Produces normal LilyPond output.

[\[Err\]](#) Causes an error or produces useless output.

Some commands and properties accept a special value:

- [EXTEXT](#): A code point, a list of code points, or markup.
- [DEFINITION](#): A string of tokens.
- [ORIENTATION](#): Sum of axis and direction.

All commands have the prefix `ekm` or `ekm-`.

## SMuFL Switches

---

```
\ekmSmuflOn ARG
\ekmSmuflOff ARG
```

Turn the SMuFL support on and off, respectively, for one or more types of graphical objects.

ARG is:

- a single symbol `TYPE`
- a list of symbols ' (`TYPE ...`)
- a list of symbols ' (`- TYPE ...`) meaning all types except for `TYPE ...`

These commands set/undo context and grob properties (usually the stencil) in the current bottom context, except for `colon` and `segno` which cannot be turned off.

Any other symbol except the following is silently ignored.

|                           |                               |
|---------------------------|-------------------------------|
| <code>all</code>          | All following types           |
| <code>systemstart</code>  | System start delimiters       |
| <code>colon</code>        | Colon bar lines               |
| <code>segno</code>        | Segno bar lines               |
| <code>clef</code>         | Clefs and clef modifiers      |
| <code>time</code>         | Time signatures               |
| <code>metronome</code>    | Metronome marks               |
| <code>notehead</code>     | Note heads                    |
| <code>dot</code>          | Augmentation dots             |
| <code>flag</code>         | Flags and grace note slashes  |
| <code>rest</code>         | Rests and multi-measure rests |
| <code>script</code>       | Scripts and expressive marks  |
| <code>dynamic</code>      | Absolute dynamic marks        |
| <code>lv</code>           | Laissez vibrer                |
| <code>tremolo</code>      | Tremolo marks                 |
| <code>arpeggio</code>     | Arpeggios                     |
| <code>percent</code>      | Percent repeats               |
| <code>textspan</code>     | Text spans                    |
| <code>trill</code>        | Trill spans and pitches       |
| <code>tuplet</code>       | Tuplet numbers                |
| <code>fingering</code>    | Fingering instructions        |
| <code>stringnumber</code> | String number indications     |
| <code>pedal</code>        | Piano pedals                  |
| <code>fbass</code>        | Figured bass                  |
| <code>lyric</code>        | Lyric text                    |
| <code>chord</code>        | Chord name symbols            |

## Example

Demonstrates possible places for SMuFL switches: a `\with` block, a `\layout` block, and in the music stream. Note that `\ekmTremolo` works independent of the `tremolo` switch which is turned on after that.

```
\new Staff \with {
  \ekmSmuflOn #'trill
}
\relative c'' {
  \ekmSmuflOn #'notehead
  \override NoteHead.style = #'triangle
  c4 a
  \ekmSmuflOff #'notehead
  \revert NoteHead.style

  \autoBeamOff
  a8
  \override Flag.style = #'straight
  a <a d> a16 <a d>
  \revert Flag.style

  \ekmPitchedTrill #'slash #'bracket
  d2 \ekmStartTrillSpan #'(-4 . 0)
  e d4 c8. a16
  \stopTrillSpan

  \ekmTremolo unmeasured
  { c4:16 a: }

  \ekmSmuflOn #'tremolo
  { c4:16 a: }
}
\layout {
  \context {
    \Score
    \ekmSmuflOn #'(flag dot)
  }
}
```



## Extended Text

---

Some commands accept an EXTEXT value, or a pair or list of EXTEXT values.  
EXTEXT can be:

- A single code point (integer). Calls `\ekm-char`.

```
#xE046
```

- A list of a single code point followed by font features:
  - One or more strings.
  - A number in the range 0 to 31 of a stylistic alternate. Higher values are treated as code points (see below).
  - A negative number to draw a path instead of the font glyph. -1 draws a filled path. Any other negative number -N draws the outline with thickness N scaled to the current font size.

Calls `\ekm-charf`.

```
' (#xE242 "salt 2")
```

```
' (#xE242 2)
```

```
' (#xE242 -1)
```

- A list of one or more code points. Calls `\ekm-chars`.

```
' (#xE262 #xE566 #xEAA6 #xEAA5)
```

- Any markup.

Note: The commands `\ekmTremolo` and `\ekmStem` interpret some strings as names of symbols.

```
"poco a poco"
```

```
(markup #:box #:ekm-char #xED19)
```

- `#f` which draws the empty-stencil.

## Definition String

---

Some commands and properties accept a DEFINITION value.

This is a string of one or more tokens, each consisting of one or more characters. Their corresponding symbols are stacked in a line. Any other character in the string produces a warning and only the text created so far is drawn. See also `\ekm-def`.

Some tokens are always applicable:

- Number tokens: A sequence of digits and / (for integer or rational number) is drawn with `\ekm-number` in a style according to the respective command (`\ekm-finger`, `\ekm-func`), or to the property `number-style`.
- Space tokens: Their default symbols emulate some Unicode whitespace.

| Token                      | Default symbol             | Unicode                   |
|----------------------------|----------------------------|---------------------------|
| <code>&lt;space&gt;</code> | <code>\hspace #1</code>    | SPACE                     |
| <code>_</code>             | <code>\hspace #0.17</code> | HAIR SPACE                |
| <code>—</code>             | <code>\hspace #0.78</code> | THIN SPACE                |
| <code>___</code>           | <code>\hspace #2</code>    | EN SPACE                  |
| <code>_____</code>         | <code>\hspace #4</code>    | EM SPACE                  |
| <code>'</code>             | <code>#f (nothing)</code>  | ZERO WIDTH NO-BREAK SPACE |

## Orientation

---

Some commands accept an ORIENTATION value.

This is the sum of

- Axis: X, Y, or  $\pm 0.5$  for diagonal.
- Direction:  $\pm 1$ , or -3 for "bilateral" orientations.

The following symbols are defined for the 12 possible values. An unsupported value is substituted with N.

| Symbol | Value | Axis | Direction |
|--------|-------|------|-----------|
| N      | 2     | Y    | UP        |
| NE     | 1.5   | 0.5  | UP        |
| E      | 1     | X    | RIGHT     |
| SE     | 0.5   | -0.5 | RIGHT     |
| S      | 0     | Y    | DOWN      |
| SW     | -0.5  | 0.5  | DOWN      |
| W      | -1    | X    | LEFT      |
| NW     | -1.5  | -0.5 | LEFT      |
| NS     | -2    | Y    | -3        |
| NESW   | -2.5  | 0.5  | -3        |
| EW     | -3    | X    | -3        |
| SENW   | -3.5  | -0.5 | -3        |

The commands `\ekm-arrow` and `\ekm-beater` support all 12 orientations. Missing musical symbols are completed by flipping or rotating. Missing bilateral musical symbols are substituted with those for N, NE, E, SE.

Note: Currently, only the arrow style `simple` used with [Ekmelos](#) has glyphs for all 12 orientations.

## Basic Markup Commands

---

These commands implement the underlying SMuFL output in Esmuflily.

`\ekm-char CODEPOINT`

Draw the glyph of CODEPOINT, or the point-stencil for zero.

Used properties:

- `font-size` (0)
- `font-name` (`ekm:font-name`)

`\ekm-charf CODEPOINT FEATURES`

Draw the glyph of CODEPOINT with font features, independent of [globally drawing paths](#).

FEATURES is:

- a list of one or more strings.
- the number of a stylistic alternate. `#1` and `#' (1)` and `#' ("salt 1")` are equivalent. `#0` and `#' ()` do not set font features.
- a negative number to draw a path instead of the font glyph. `#-1` and `#' (-1)` draw a filled path. Any other negative number `-N` draws the outline with thickness `N` scaled to the current font size.

Used properties:

- `font-size` (0)
- `font-name` (`ekm:font-name`)

Examples:

`\ekm-charf ##xE242 #0`



`\ekm-charf ##xE242 #' ("salt 1")`



`\ekm-charf ##xE242 #' (2)`



`\ekm-charf ##xE242 #-20`



`\ekm-str STRING`

Draw STRING, independent of [globally drawing paths](#).

Used properties:

- `font-size` (0)
- `font-name` (`ekm:font-name`)

`\ekm-chars CODEPOINT-LIST`

Draw the glyphs of the CODEPOINTS in the list adjoined horizontally without padding, or the point-stencil for an empty list.

Used property:

- `font-size` (0)

Examples:

`\ekm-chars #' (#xE260 #xE2B4 #xE2B2)`



`\ekm-chars #' (#xE262 #xE566 #xEAA6 #xEAA5)`



`\ekm-chars #' (#xE1F0 #xE1F7 #xE1FC #xE1F7 #xE1F4)`



`\ekm-text EXTEXT`

Draw **EXTEXT**. Depending on the argument type, it calls `\ekm-char`, `\ekm-charf`, or `\ekm-chars`, or it draws markup or the empty-stencil.

Examples:

`\ekm-text #'(#xE242 0)`



`\ekm-text #'(#xE242 -20)`



`\ekm-text #'(#xE260 #xE2B4 #xE2B2)`



`\ekm-concat EXTEXT-LIST`

Draw the **EXTEXT**s in the list stacked in a line without padding.

`\ekm-line EXTEXT-LIST`

Draw the **EXTEXT**s in the list stacked in a line.

Used properties:

- `word-space`
- `text-direction`

Examples:

`\ekm-line #'(#xE046 "al fine")`

D.C. al fine

`\ekm-line #'((#xE6D0 1) "with" #xE78E)`



`\ekm-join-line SEPARATOR ARGS`

Draw the markups in the list **ARGS** with the markup **SEPARATOR** in between (infix) stacked in a line.

Used properties:

- `word-space`
- `text-direction`

`\ekm-combine CODEPOINT X Y CODEPOINT2`

Combine the glyphs of **CODEPOINT** and **CODEPOINT2**, where **CODEPOINT2** is translated scaled by **X,Y**.

Examples:

`\ekm-combine ##xECA5 #-0.5 #1.0 ##xE56E`



`\ekm-combine ##xEA7F #0.3 #0 ##xE87B`



`\ekm-cchar CENTER CODEPOINT`

Draw the glyph of **CODEPOINT**, centered horizontally if **CENTER** is **CX** or **CXY**, and vertically if **CENTER** is **CY** or **CXY**.

`\ekm-ctext CENTER EXTEXT`

Draw **EXTEXT** centered like `\ekm-cchar`.

If **CENTER** is **STEMCX** or **STEMCXY** (intended to draw symbols on stem) then:

- a list of code points is centered only horizontally.
- a single code point (possibly with font features) is never centered.

`\ekm-number` STYLE NUMBER

Draw NUMBER (a number or string) according to STYLE (a symbol), either composed of decimal digit symbols, or a precomposed number symbol (\*), or created with a markup procedure.

If there is no precomposed number symbol defined for NUMBER, it draws the digits from another style which can be overridden with property `fall-back-style`.

A non-integer number is drawn either as a common fraction or with integer and fraction part if property `mixed` is true. A plus sign is inserted if `mixed` is 'plus.

Numerator and denominator of a fraction are aligned according to property `fraction-style`:

- 'default or 'vertical: Vertical with a horizontal bar and `y-padding`.
- 'none: Vertical with `y-padding` without a bar.
- 'horizontal or 'line: Horizontal with a solidus and `x-padding`.
- 'diagonal or 'slash: Oblique with a diagonal bar and `xy-padding`, or a precomposed fraction.

Used properties:

- `font-size` (0)
- `mixed` (#t)
- `fraction-style` ('default)
- `fraction-size` (-6) relative to the font size.
- `fraction-align` (DOWN)
- `bar-thickness` (0.15)
- `x-padding` (0.2)
- `xy-padding` (0.1)
- `y-padding` (0.1)
- `word-space`
- `fall-back-style` ('())

## STYLE

|                   |                                                                                     |                                                                 |
|-------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 'default          | 4                                                                                   | Uses a LilyPond font, see <code>\ekm-lily</code> further below. |
| 'sans             | 4                                                                                   | -"-                                                             |
| 'serif            | 4                                                                                   | -"-                                                             |
| 'typewriter       | 4                                                                                   | -"-                                                             |
| 'time             |  | Time signature U+E080-U+E089                                    |
|                   |  | Precomposed fraction U+E097-U+E09B                              |
| 'time-turned      |  | Turned time signature U+ECE0-U+ECE9                             |
| 'time-reversed    |  | Reversed time signature U+ECF0-U+ECF9                           |
| 'tuplet           |  | Tuplet number U+E880-U+E889                                     |
| 'fingering        |  | Fingering U+ED10-U+ED15, U+ED24-U+ED27                          |
| 'fingering-italic |  | Fingering italic U+ED80-U+ED89                                  |
| 'clef-modifier    | 8                                                                                   | Clef modifier U+E07D, U+E07E or 'fingering-italic (*)           |
| 'fbass            | 4                                                                                   | Figured bass U+EA50 .. U+EA61                                   |
| 'func             |  | Function theory U+EA70-U+EA79                                   |
| 'string           |  | Guitar string U+E833-U+E83C, U+E84A-U+E84D or 'sans (*)         |
| 'scale            |  | Scale degree U+EF00-U+EF08 or 'sans (*)                         |

`\ekm-with-parens USE-STYLE STYLE ARG`

Draw markup ARG with parentheses according to USE-STYLE (a symbol) and STYLE (a symbol).

If a USE-STYLE draws symbols from another style (\*), this can be overridden with property `fall-back-style`.

Used property:

- `fall-back-style (')`

## USE-STYLE

|                                |                                                                 |
|--------------------------------|-----------------------------------------------------------------|
| <code>'default</code>          | Uses a LilyPond font, see <code>\ekm-lily</code> further below. |
| <code>'sans</code>             | <code>-"</code>                                                 |
| <code>'serif</code>            | <code>-"</code>                                                 |
| <code>'typewriter</code>       | <code>-"</code>                                                 |
| <code>'dynamic</code>          | Basic Latin symbols with padding                                |
| <code>'hairpin</code>          | U+E542-U+E545                                                   |
| <code>'clef-modifier</code>    | <code>'fingering-italic (*)</code>                              |
| <code>'fingering</code>        | U+ED28-U+ED2B                                                   |
| <code>'fingering-italic</code> | U+ED8A-U+ED8D                                                   |
| <code>'time</code>             | U+E092-U+E093, U+EC82-U+EC83                                    |
| <code>'chord</code>            | U+E875-U+E878                                                   |
| <code>'fbass</code>            | U+EA68-U+EA6B                                                   |
| <code>'func</code>             | U+EA8F-U+EA94                                                   |
| <code>'notehead</code>         | U+E0F5-U+E0F6                                                   |
| <code>'accidental</code>       | U+E26A-U+E26D                                                   |

## STYLE

|                             |                          |
|-----------------------------|--------------------------|
| <code>'default</code>       | <code>(...)</code>       |
| <code>'parens</code>        | <code>(...)</code>       |
| <code>'parenthesized</code> | <code>(...)</code>       |
| <code>'bracket</code>       | <code>[...]</code>       |
| <code>'bracketed</code>     | <code>[...]</code>       |
| <code>'brace</code>         | <code>{...}</code>       |
| <code>'angle</code>         | <code>&lt;...&gt;</code> |

`\ekm-lily STYLE EXTEXT`

Draw EXTEXT with a LilyPond font according to STYLE (a symbol).

Note: This command cannot draw [paths](#).

## STYLE

|                          |                                                                      |
|--------------------------|----------------------------------------------------------------------|
| <code>'default</code>    | Calls <code>\simple</code>                                           |
| <code>'sans</code>       | Calls <code>\sans</code>                                             |
| <code>'serif</code>      | Calls <code>\roman</code> in LilyPond 2.24, else <code>\serif</code> |
| <code>'typewriter</code> | Calls <code>\typewriter</code>                                       |

`\ekm-def MAP DEFINITION`

Draw a text according to [DEFINITION](#).

MAP is an alist of [EXTTEXT](#)s mapped onto tokens (strings), or a type name (a symbol). A token which is a prefix of other tokens must be arranged after them in MAP. So the correct order is "abc", "ab", "a". The tokens in MAP take precedence over [number and space tokens](#). The special value `#f` draws nothing, i.e. the token is simply ignored.

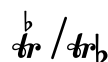
Used property:

- `number-style (#f)`, see [\ekm-number](#)

Examples:

```
#(define my-map `(
  ("beater/" . , (markup
    #:ekm-beater 'timpani-hard NE))
  ("b" . #xE260)
  ("trb" . (#xE260 #xE566))
  ("tr" . #xE566)
  ("timp" . (#xE6D0 1))
  ("/" . , (markup
    #:override '(thickness . 2)
    #:draw-line '(1 . 3)))
))
```

```
\ekm-def #my-map #"trb__/_tr'b"
```



```
\ekm-def #my-map #"timpbeater//b"
```



`\ekm-label ORIENTATION LABEL ARG`

Combine markup ARG with another markup LABEL placed next to it according to [ORIENTATION](#), where `#f` ignores the label.

Used properties:

- `font-size (0)`
- `label-size (-4)` relative to the font size.
- `padding (0.3)`

Examples:

```
\ekm-label #SE
  \ekm-char ##xE836
  "G"
```



```
\ekm-label #NW
  \ekm-beater #'timpani-hard #NE
  \ekm-char ##xE802
```



`\ekm-orient TYPE STYLE ORIENTATION`

Draw the symbol of TYPE and STYLE according to [ORIENTATION](#) as markup.

Examples:

```
\ekm-orient #'arrow #'black #NW
```



```
\ekm-orient #'beater #'timpani-medium #NE
```



`\ekm-script STYLE DIRECTION`

Draw the musical symbol of TYPE and STYLE according to DIRECTION as markup, where TYPE is:

- `'script` if STYLE is a string.
- `'spanner` and the edge symbol of the spanner is drawn if STYLE is a symbol.

This command is intended to combine symbols with accidentals.

Examples:

`\ekm-script #"dmарcato" #UP`

**^**

`\ekm-script #'trill #UP`

**tr**

## Cosmuflily

---

Cosmuflily = Esmuflily + Ekmelily

Combine Esmuflily with [Ekmelily](#) which supports accidentals and key signatures in several tunings, for a full [SMuFL](#) support.

### Usage

Install from the Ekmelily folder [ly](#) the file for the desired tuning, e.g. `ekmel-24.ily`, and the main include file `ekmel-main.ily`, as well as the [Esmuflily files](#) together with `cosmufl.ily`.

Add the following lines near the top of your LilyPond input file.

`ekmFont` and `ekmUse` are optional but must be placed before `\include`.

```
ekmFont = "FONT"
ekmUse = USE
\include "cosmufl.ily"
```

USE is a sequence of tuning, notation style, and language (each is optional), followed by any number of SMuFL [switches](#), either as a string separated by whitespace, or as a list of symbols or strings.

Tuning can also be a number. It must be the first item if any. The default is 24 (includes `ekmel-24.ily`).

Style and language are recognized only if defined in the selected tuning. The language name takes precedence in case of identical names (`arabic` in tuning "arabic" and `thm` in tuning 53). The default language is `nederlands` in most tunings. The default style is `stc` (Stein/Couper) in tuning 24.

The SMuFL [switches](#) can be preceded by a `+`, or by a `-` which turns on all but the specified switches. The default is `all` (equivalent to a sole `-`).

`cosmufl.ily` includes `esmufl.ily` and the Ekmelily file for the selected tuning, and calls `\ekmStyle`, `\language`, and `\ekmSmuflOn` (in the Score context).

### USE Examples:

- Set tuning 31, Standard style, language `nederlands`, and all switches.

```
ekmUse = 31
```

- Set tuning 24, Stein/Couper style, language `english`, and all switches.

```
ekmUse = "english"
```

- Set tuning 5-limit JI, Extended Helmholtz-Ellis style, language `nederlands`, and all switches.

```
ekmUse = "5JI he"
```

- Set tuning 24, Gould style, language `italiano`, and switch for note heads.

```
ekmUse = #'(italiano go notehead)
```

- Set tuning 72, Sims style, language `svenska`, and all switches except for colon and segno bar lines.

```
ekmUse = #'(72 sims svenska - colon segno)
```

## Cosmuflily Commands

---

`cosmuflily` defines the following additional commands.

`\ekm-tuning`

Draw the selected tuning as markup.

`\ekm-combine-accidental ALTERATION-1 ALTERATION-2 ARG`

Combine markup ARG with the accidentals of ALTERATION-1 and 2 according to the selected notation style.

ALTERATION is a rational number, an alteration name (see [Ekmelily](#)), or 'none which draws no accidental (empty-stencil).

If ALTERATION-2 is 'x the accidental of ALTERATION-1 is placed to the right of ARG with `x-padding`. Else the accidentals are placed centered above and below ARG with `y-padding`.

Used properties:

- `x-padding` (0.2)
- `y-padding` (0.1)
- `acc-size` (-1) relative to the font size.

`\ekmScriptAccidental SCRIPT-NAME ALTERATION-1 ALTERATION-2`

Create a script from SCRIPT-NAME combined with the accidentals of ALTERATION-1 and 2.

Note: This does not work if SCRIPT-NAME is different from the internal name, such as `marcato` and `dmarcato`.

See also [Scripts and Expressive Marks](#).

`\ekmStartTrillSpanAccidental TEMPO ALTERATION`

Start a trill span whose left edge symbol is combined with the accidental of ALTERATION. [\[LP\]](#)

See also [Trill spans and pitches](#).

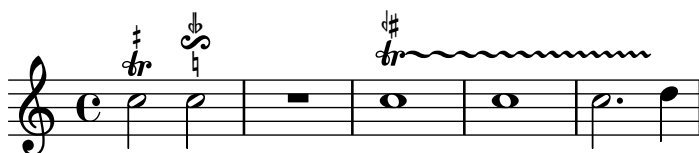
## Example

Uses the default tuning 24 and language `nederlands`. Selects the Stein/Zimmermann notation style (`stz`), and turns on the SMuFL support for scripts and trill spans. The `+` is optional, since the SMuFL switches won't be recognized as language names.

```
ekmUse = #'(stz + script trill)
\include "cosmufl.ily"

\relative c'' {
  c2 ^ \ekmScriptAccidental #'trill #1/4 #'none
  c2 ^ \ekmScriptAccidental #'reverseturn #-3/4 #0
  R1

  c1 \ekmStartTrillSpanAccidental
    #'(-3 . 2)
    #'iseh
  c c2. d4
  \stopTrillSpan
}
```



## Musical Symbols

---

A musical symbol supported by Esmuflily is defined with an [EXTEXT](#) value. This is

- a code point,
- a list of code points,
- or a markup.

All default musical symbols are SMuFL recommended characters defined with their Unicode code point. See the Ekmelos include file [ekmelos-map.ily](#) with definitions to access glyphs by name.

In this document, some font-specific symbols of [Ekmelos](#) or [Bravura](#) are listed after the default symbols.

### Maintenance

Esmuflily maintains the musical symbols in hierarchical categories:

- The main category is the type. Most of the types correspond to LilyPond's graphical objects, like note heads, flags, rests, and clefs.
- Sub-categories are style, duration log, direction, name (identifier), token (for definition strings), maximum size, and others.

### Examples

- $\infty$  (noteheadXHalf, U+E0A8) is associated with type `notehead`, style `cross`, and duration log 1.
- $\}$  (flag8thUp, U+E240) is associated with type `flag`, style `default`, duration log 3, and direction `UP`.
- $\text{Sost.}$  (keyboardPedalSost, U+E659) is associated with type `pedal`, and token `"Sost."`.

### Customize musical symbols

Note: All musical symbols are assembled in a single internal table.

The style `ekm` is reserved for special purpose.

```
\ekmMergeType TYPE TABLE
```

Merge `TABLE` (an alist) with musical symbols for `TYPE` (a symbol) into the internal table, i.e. new musical symbols, styles, names, and tokens are added, and already existing ones are replaced.

See [Internals](#) at Wiki for all types and the table structure.

If `TYPE` is `'defaults` or `'glyphs`, `TABLE` is merged into the respective metadata table.

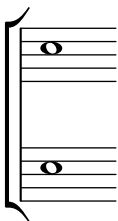
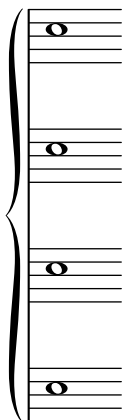
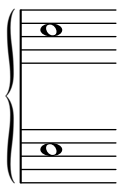
See [Internals of Metadata](#) at Wiki.

## System Start Delimiters

---

`\ekmSmuflOn #'systemstart`

Draw SMuFL system start delimiters, using `\ekm-system-start`.



```
\ekm-system-start STYLE HEIGHT
```

Draw a system start delimiter of HEIGHT in staff units as markup.

Used property:

- `font-size (0)`

STYLE

|          |   |        |               |
|----------|---|--------|---------------|
| 'brace   | } | U+E000 | brace         |
| 'bracket | ⌒ | U+E003 | bracketTop    |
|          | ⌓ | U+E004 | bracketBottom |

with Bravura or Ekmelos

'brace makes use of Bravura's stylistic alternates or Ekmelos' size variants, each intended for a specific size range.

## Staff Dividers and Separators

---

`\ekmStaffDivider DIRECTION`

Draw the next barline with an indicator to split or recombine the staff and set a `\break`.

**DIRECTION**

|        |                                                                                   |        |                        |
|--------|-----------------------------------------------------------------------------------|--------|------------------------|
| DOWN   |  | U+E00B | staffDivideArrowDown   |
| UP     |  | U+E00C | staffDivideArrowUp     |
| CENTER |  | U+E00D | staffDivideArrowUpDown |

`system-separator-markup = \ekmSlashSeparator SIZE`

Draw a system separator mark, corresponding to **SIZE** (set within a `\paper block`).

**SIZE**

|          |                                                                                   |        |                        |
|----------|-----------------------------------------------------------------------------------|--------|------------------------|
| 0        |  | U+E007 | systemDivider          |
| $\leq 1$ |  | U+E008 | systemDividerLong      |
| $> 1$    |  | U+E009 | systemDividerExtraLong |

### Example

```
\new Staff
<<
  \new Voice {
    \relative c'' {
      \voiceOne
      g4 a b c
      \bar "||" \ekmStaffDivider #CENTER
    }
  }
  \new Voice {
    \relative c' {
      \voiceTwo
      e4 c f e
    }
  }
>>
```



## Colon and Segno Bar Lines

---

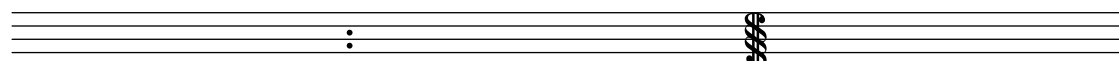
```
\ekmSmuflOn #'colon
```

```
\ekmSmuflOn #'segno
```

Draw SMuFL colon and segno bar lines, respectively.

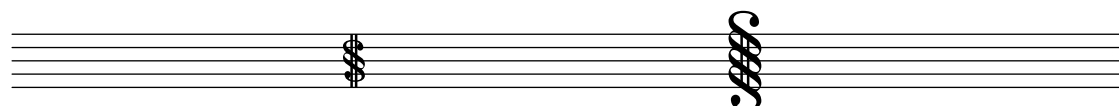
Note: Both [switches](#) are set independently of a context and cannot be turned off.

|   |        |               |
|---|--------|---------------|
| : | U+E043 | repeatDots    |
| S | U+E04A | segnoSerpent1 |



with Ekmelos

|    |        |                    |
|----|--------|--------------------|
| S  | U+F6C8 | segnoSerpentSmall1 |
| \$ | U+F6CA | segnoSerpentLarge1 |



## Examples

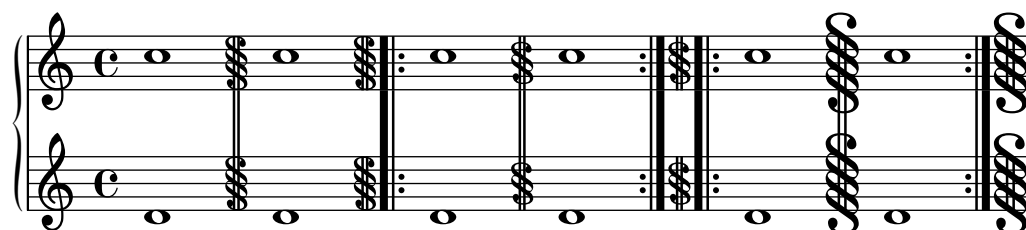
```

\new PianoStaff \with {
  \ekmSmuflOn #'segno
}
<<
  \new Staff \relative c'' {
    c1 \bar "S"
    c \bar "S. | :-S"

    % Ekmelos bar glyphs
    c \bar "s"
    c \bar ": | .s. | :-s"

    c \bar "$"
    c \bar ": | .s-$"
  }
  \new Staff \relative c' {
    d1 d d d d d
  }
>>

```



Clefs

```
\ekmSmuflOn #'clef
```

Draw SMuFL clefs and clef modifiers.

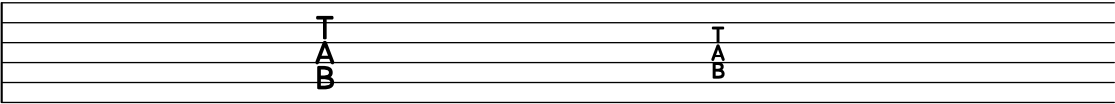
|                           |        |               |
|---------------------------|--------|---------------|
| "G" "treble" "french" ... | U+E050 | gClef         |
| "GG"                      | U+E055 | gClef8vbOld   |
| "tenorG"                  | U+E056 | gClef8vbCClef |
| "F" "bass" ...            | U+E062 | fClef         |
| "C" "alto" ...            | U+E05C | cClef         |
| "neomensural-c1" ...      | U+E060 | cClefSquare   |



|                  |        |                            |
|------------------|--------|----------------------------|
| "bridge"         | U+E078 | bridgeClef                 |
| "accordion"      | U+E079 | accdnDiatonicClef          |
| "percussion"     | U+E069 | unpitchedPercussionClef1   |
| "varpercussion"  | U+E06A | unpitchedPercussionClef2   |
| "semipitched"    | U+E06B | semipitchedPercussionClef1 |
| "varsemipitched" | U+E06C | semipitchedPercussionClef2 |
| "indiandrum"     | U+ED70 | indianDrumClef             |



|              |        |                |
|--------------|--------|----------------|
| "tab"        | U+E06D | 6stringTabClef |
| "4stringtab" | U+E06E | 4stringTabClef |



## Change clefs

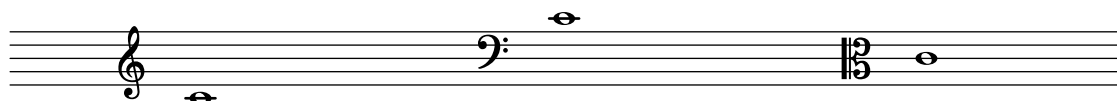
They are drawn either with a special glyph or with the normal glyph but smaller.

The relative font size for change clefs can be set with

```
#(set! ekm:clef-change-font-size '(SPECIAL-SIZE . NORMAL-SIZE))
```

The standard value is '(1.5 . -2).

|     |        |             |
|-----|--------|-------------|
| "G" | U+E07A | gClefChange |
| "F" | U+E07C | fClefChange |
| "C" | U+E07B | cClefChange |



## Clef modifiers

```
\ekmLigaClefsOn  
\ekmLigaClefsOff
```

Turn the ligated mode for normal, change, and cue clefs on and off, respectively.

In the ligated mode, a clef is drawn including the modifier, either a precomposed glyph or the modifier is combined with a clef which is either a special glyph for ligating or the pure clef glyph. Else, clef and modifier are drawn separately as usual in LilyPond.

Transposition and style are drawn with the fingering italic symbols for digits, parentheses, and brackets, and with the following special symbols.

|    |           |        |        |
|----|-----------|--------|--------|
| 8  | <i>8</i>  | U+E07D | clef8  |
| 15 | <i>15</i> | U+E07E | clef15 |

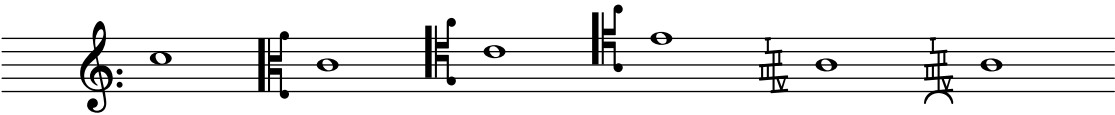
```
"G_8"  
"G_(15)"  
"GG^[4]"  
"F_(3)"  
"subbass^13"  
"C_2"
```



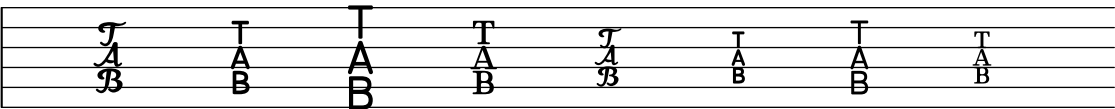


with Ekmelos

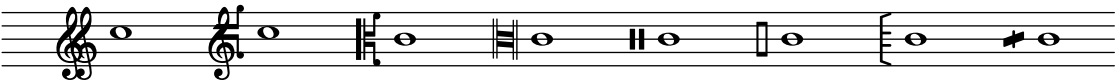
|                      |        |                        |
|----------------------|--------|------------------------|
| "frenchG"            | U+F40E | gClef8vbFrench         |
| "varC" "altovarC"    | U+F633 | cClefFrench20C         |
| "tenorvarC"          | :      |                        |
| "baritonevarC"       | :      |                        |
| "string"             | U+F71C | stringClef             |
| "behindbridgestring" | U+F71D | behindBridgeStringClef |



|                    |        |                       |
|--------------------|--------|-----------------------|
| "tab"              | U+F61E | 6stringTabClefClassic |
| "moderntab"        | U+E06D | 6stringTabClef        |
| "talltab"          | U+F40A | 6stringTabClefTall    |
| "seriftab"         | U+F40B | 6stringTabClefSerif   |
| "4stringtab"       | U+F61F | 4stringTabClefClassic |
| "4stringmoderntab" | U+E06E | 4stringTabClef        |
| "4stringtalltab"   | U+F40C | 4stringTabClefTall    |
| "4stringseriftab"  | U+F40D | 4stringTabClefSerif   |



|                  |        |                                  |
|------------------|--------|----------------------------------|
| "GG"             | U+F630 | gClef8vbOldChange                |
| "tenorG"         | U+F631 | gClef8vbCClefChange              |
| "varC"           | U+F634 | cClefFrench20CChange             |
| "neomensural-c3" | U+F632 | cClefSquareChange                |
| "percussion"     | U+F635 | unpitchedPercussionClef1Change   |
| "varpercussion"  | U+F636 | unpitchedPercussionClef2Change   |
| "semipitched"    | U+F6BE | semipitchedPercussionClef1Change |
| "varsemipitched" | U+F6BF | semipitchedPercussionClef2Change |



## Clef Markup

---

`\ekm-clef NAME MODE`

Draw a clef with or without a modifier according to NAME (a string) and MODE (a symbol) as markup.

NAME is one of:

- "GLYPHNAME"
- "GLYPHNAME\_change"
- "GLYPHNAME\_TRANSPOSITION\_STYLE"
- "GLYPHNAME\_TRANSPOSITION\_STYLE\_change"

GLYPHNAME is the glyph name of the clef, e.g. `clefs.G` (not `G`).

STYLE is the parenthesis style of the modifier, see `\ekm-clef-modifier`.

MODE is the ligated mode:

- 'default ignores TRANSPOSITION and STYLE in NAME and draws a pure clef glyph.
- 'liga draws a ligated clef.

`\ekm-clef-modifier TRANSPOSITION STYLE`

Draw a clef modifier with TRANSPOSITION (a number or string) and STYLE (a symbol) as markup.

STYLE

|                |                |
|----------------|----------------|
| 'default       | No parentheses |
| 'parenthesized | (...)          |
| 'bracketed     | [...]          |

## Time Signatures

---

`\ekmSmuflOn #'time`

Draw SMuFL time signatures.

Note: It supports fractional and complex time signatures, and the Scheme syntax as introduced in LilyPond 2.26. Esmuflily's own previous syntax is still available for older LilyPond.

See `\ekmCompoundMeter` further below.

`\ekm-compound-meter TIME-SIGNATURE`

Draw a numeric time signature as markup. Sets a large plus between fractions, and a small plus within numerators.

It provides some additional property values:

- `denominator-style`: 'diagonal or 'slash draws fractions with a large slash.
- `nested-fraction-orientation`: 'diagonal or 'slash draws nested fractions with a slash or as a precomposed glyph.
- `note-head-style`: Any style of a [note head](#) or of a [precomposed note](#).

Used properties:

- `font-size` (0)
- `denominator-style` ('default)
- `nested-fraction-mixed` (#t)
- `nested-fraction-orientation` ('default)
- `nested-fraction-relative-font-size` ('())
- `note-dots-direction` (CENTER)
- `note-flag-style` ('())
- `note-head-style` ('metronome)
- `note-staff-position` (-2)
- `bar-thickness` (0.25) used if no slash glyph is available
- `xy-padding` (0.1)

|       |                                                                                     |        |                        |
|-------|-------------------------------------------------------------------------------------|--------|------------------------|
| 4 / 4 |  | U+E08A | timeSigCommon          |
| 2 / 2 |  | U+E08B | timeSigCutCommon       |
| plus  |  | U+E08C | timeSigPlus            |
|       |  | U+E08D | timeSigPlusSmall       |
| slash |  | U+E08E | timeSigFractionalSlash |
|       |  | U+EC84 | timeSigSlash           |

See [\ekm-number](#) for time signature digits and precomposed fractions, and for the properties `bar-thickness` and `xy-padding`.

```
\ekmCompoundMeter TIME-SIGNATURE
```

Set a numeric time signature. A numerator can contain nested fractions for the SMuFL precomposed glyphs. If specified in a pair, e.g.  $(1 \ . \ 1/2)$ , this is treated as a single number without a plus sign.

Note: This function is deprecated and should be used only with LilyPond prior to 2.26. The Scheme syntax of TIME-SIGNATURE, the style of nested fractions, and the beaming behavior are different. The same output as with

```
\ekmCompoundMeter #'((2 4) ((1 . 1/2) 4))
```

can be achieved for LilyPond 2.26 with

```
\override Timing.TimeSignature.nested-fraction-orientation =
  #'diagonal
\time #'((2 . 4) (3/2 . 4))
\set Timing.beamExceptions = #'()
\set Timing.beatBase = #1/8
\set Timing.beatStructure = 4,3
```

## Examples

```

\new Staff \with {
  \ekmSmuflOn #'time
}
\relative c'' {
  \time 5/8
  \repeat unfold 5 c8

  \time #'((1 . 4) (3 . 8))
  \repeat unfold 5 c8

  \once \override Timing.TimeSignature.denominator-style =
    #'diagonal
  \time #'((1 . 4) (3 . 8))
  \repeat unfold 5 c8

  \time #'((2 . 4) (3/2 . 4))
  \repeat unfold 7 c8

  \override Timing.TimeSignature.nested-fraction-orientation =
    #'horizontal
  \time #'(5/2 . 4)
  c2 r8

  \override Timing.TimeSignature.nested-fraction-orientation =
    #'diagonal
  \time #'(5/2 . 4)
  c2 r8

  \override Timing.TimeSignature.denominator-style =
    #'note
  \time #'(((1 2 3) . 8) (2 . 8/3))
  \repeat unfold 12 c8
}

```



# Cadenza Signatures

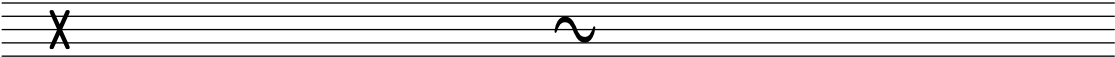
---

\ekmCadenzaOn STYLE

Start a cadenza like \cadenzaOn and set a SMuFL signature.

STYLE

|     |        |                       |
|-----|--------|-----------------------|
| "X" | U+E09C | timeSigX              |
| "~" | U+E09D | timeSigOpenPenderecki |

A musical staff with five lines. On the left, there is a large, bold 'X' symbol. On the right, there is a tilde '~' symbol. Both symbols are positioned between the second and third lines of the staff.

## Metronome Marks

---

```
\ekmSmuflOn #'metronome
```

Draw SMuFL notes in metronome marks specified with `\tempo`, using `\ekm-note-by-number`.

The default style of the notes is `'metronome`. Another style can be set with

```
\override Score.MetronomeMark.style = #STYLE
```

It can be any style of a [note head](#) or of a [precomposed note](#). In the latter case, no extra stem or flag is drawn and the property `flag-style` is ignored.

The direction is always UP.

## Note Heads

`\ekmSmuflOn #'notehead`

Draw SMuFL note heads. The `harmonic` and `cross` glyphs are also used with commands like `\harmonic` and `\xNote`.

The style can be set with `\override NoteHead.style = #STYLE`

### STYLE

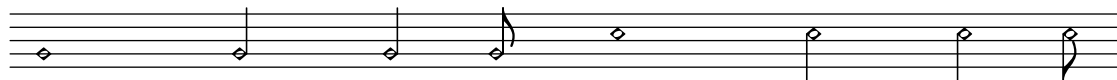
|                       |                     |                                  |
|-----------------------|---------------------|----------------------------------|
| <code>'default</code> | <code>U+E0A0</code> | <code>noteheadDoubleWhole</code> |
|                       | <code>U+E0A2</code> | <code>noteheadWhole</code>       |
|                       | <code>U+E0A3</code> | <code>noteheadHalf</code>        |
|                       | <code>U+E0A4</code> | <code>noteheadBlack</code>       |



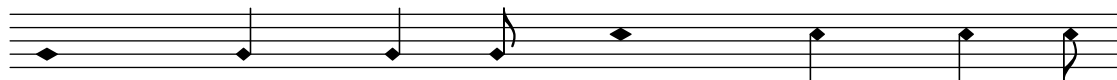
|                       |                     |                                        |
|-----------------------|---------------------|----------------------------------------|
| <code>'baroque</code> | <code>U+E0A1</code> | <code>noteheadDoubleWholeSquare</code> |
|                       | :                   |                                        |



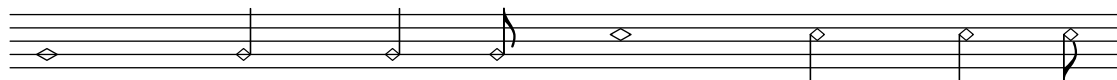
|                        |                     |                                  |
|------------------------|---------------------|----------------------------------|
| <code>'harmonic</code> | <code>U+E0D9</code> | <code>noteheadDiamondHalf</code> |
|------------------------|---------------------|----------------------------------|



|                              |                     |                                       |
|------------------------------|---------------------|---------------------------------------|
| <code>'harmonic-black</code> | <code>U+E0DC</code> | <code>noteheadDiamondBlackWide</code> |
|                              | <code>U+E0DB</code> | <code>noteheadDiamondBlack</code>     |



|                              |                     |                                       |
|------------------------------|---------------------|---------------------------------------|
| <code>'harmonic-white</code> | <code>U+E0DE</code> | <code>noteheadDiamondWhiteWide</code> |
|                              | <code>U+E0DD</code> | <code>noteheadDiamondWhite</code>     |



|                              |                     |                                         |
|------------------------------|---------------------|-----------------------------------------|
| <code>'harmonic-mixed</code> | <code>U+E0D7</code> | <code>noteheadDiamondDoubleWhole</code> |
|                              | <code>U+E0D8</code> | <code>noteheadDiamondWhole</code>       |
|                              | <code>U+E0D9</code> | <code>noteheadDiamondHalf</code>        |
|                              | <code>U+E0DB</code> | <code>noteheadDiamondBlack</code>       |



|                             |                     |                                         |
|-----------------------------|---------------------|-----------------------------------------|
| <code>'harmonic-wide</code> | <code>U+E0D7</code> | <code>noteheadDiamondDoubleWhole</code> |
|                             | <code>U+E0D8</code> | <code>noteheadDiamondWhole</code>       |
|                             | <code>U+E0DA</code> | <code>noteheadDiamondHalfWide</code>    |
|                             | <code>U+E0DC</code> | <code>noteheadDiamondBlackWide</code>   |



'diamond

U+E0DF noteheadDiamondDoubleWholeOld  
 U+E0E0 noteheadDiamondWholeOld  
 U+E0E1 noteheadDiamondHalfOld  
 U+E0E2 noteheadDiamondBlackOld



'cross

U+E0A6 noteheadXDoubleWhole  
 U+E0A7 noteheadXWhole  
 U+E0A8 noteheadXHalf  
 U+E0A9 noteheadXBlack



'xcircle

U+E0B0 noteheadCircleXDoubleWhole  
 U+E0B1 noteheadCircleXWhole  
 U+E0B2 noteheadCircleXHalf  
 U+E0B3 noteheadCircleX



'withx

U+E0B4 noteheadDoubleWholeWithX  
 U+E0B5 noteheadWholeWithX  
 U+E0B6 noteheadHalfWithX  
 U+E0B7 noteheadVoidWithX



'plus

U+E0AC noteheadPlusDoubleWhole  
 U+E0AD noteheadPlusWhole  
 U+E0AE noteheadPlusHalf  
 U+E0AF noteheadPlusBlack



'slashed

U+E0D5 noteheadSlashedDoubleWhole1  
 U+E0D3 noteheadSlashedWhole1  
 U+E0D1 noteheadSlashedHalf1  
 U+E0CF noteheadSlashedBlack1



'backslashed

U+E0D6 noteheadSlashedDoubleWhole2  
 U+E0D4 noteheadSlashedWhole2  
 U+E0D2 noteheadSlashedHalf2  
 U+E0D0 noteheadSlashedBlack2



'triangle

|        |                                 |
|--------|---------------------------------|
| U+E0BA | noteheadTriangleUpDoubleWhole   |
| U+E0BB | noteheadTriangleUpWhole         |
| U+E0BC | noteheadTriangleUpHalf          |
| U+E0BE | noteheadTriangleUpBlack         |
| U+E0C3 | noteheadTriangleDownDoubleWhole |
| U+E0C4 | noteheadTriangleDownWhole       |
| U+E0C5 | noteheadTriangleDownHalf        |
| U+E0C7 | noteheadTriangleDownBlack       |



'triangle-up



'triangle-down



'arrow

|        |                                   |
|--------|-----------------------------------|
| U+E0ED | noteheadLargeArrowUpDoubleWhole   |
| U+E0EE | noteheadLargeArrowUpWhole         |
| U+E0EF | noteheadLargeArrowUpHalf          |
| U+E0F0 | noteheadLargeArrowUpBlack         |
| U+E0F1 | noteheadLargeArrowDownDoubleWhole |
| U+E0F2 | noteheadLargeArrowDownWhole       |
| U+E0F3 | noteheadLargeArrowDownHalf        |
| U+E0F4 | noteheadLargeArrowDownBlack       |



'arrow-up



'arrow-down



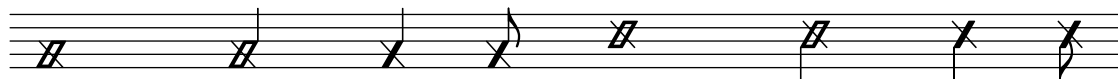
'slash

U+E10A noteheadSlashWhiteDoubleWhole  
 U+E102 noteheadSlashWhiteWhole  
 U+E103 noteheadSlashWhiteHalf  
 U+E101 noteheadSlashHorizontalEnds



'slash-muted

U+E109 noteheadSlashWhiteMuted  
 U+E108 noteheadSlashHorizontalEndsMuted



'circled

U+E0E7 noteheadCircledDoubleWhole  
 U+E0E6 noteheadCircledWhole  
 U+E0E5 noteheadCircledHalf  
 U+E0E4 noteheadCircledBlack



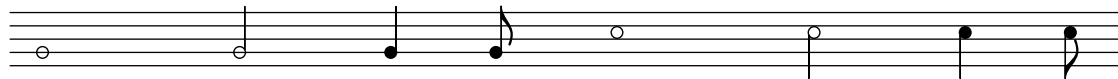
'circled-large

U+E0EB noteheadCircledDoubleWholeLarge  
 U+E0EA noteheadCircledWholeLarge  
 U+E0E9 noteheadCircledHalfLarge  
 U+E0E8 noteheadCircledBlackLarge



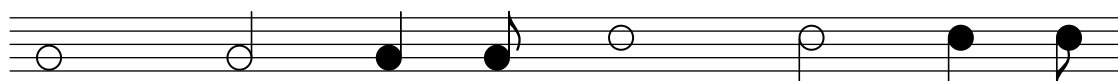
'round

U+E114 noteheadRoundWhite  
 U+E113 noteheadRoundBlack



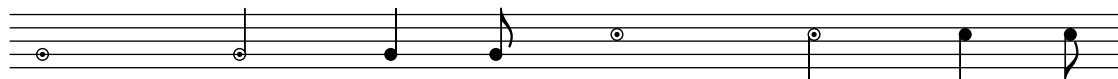
'round-large

U+E111 noteheadRoundWhiteLarge  
 U+E110 noteheadRoundBlackLarge



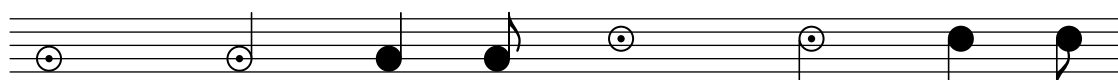
'round-dot

U+E115 noteheadRoundWhiteWithDot  
 U+E113 noteheadRoundBlack



'round-dot-large

U+E112 noteheadRoundWhiteWithDotLarge  
 U+E110 noteheadRoundBlackLarge



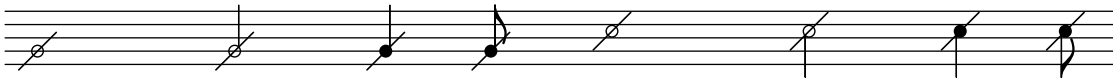
'round-slashed

U+E119

noteheadRoundWhiteSlashed

U+E118

noteheadRoundBlackSlashed



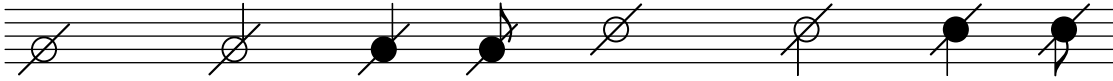
'round-slashed-large

U+E117

noteheadRoundWhiteSlashedLarge

U+E116

noteheadRoundBlackSlashedLarge



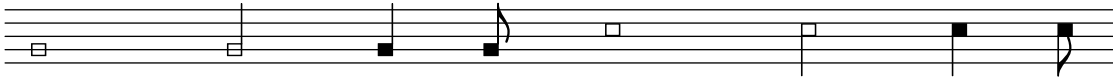
'square

U+E0B8

noteheadSquareWhite

U+E0B9

noteheadSquareBlack



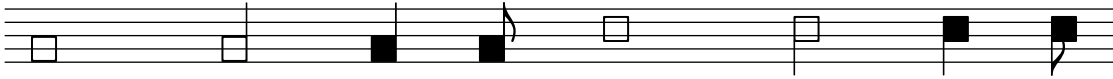
'square-large

U+E11B

noteheadSquareBlackWhite

U+E11A

noteheadSquareBlackLarge



with Ekmelos

```
'default
```

```
U+F637    noteheadLongaUp
U+F638    noteheadLongaDown
:
```



```
'altdefault
```

|        |                        |
|--------|------------------------|
| U+F637 | noteheadLongaUp        |
| U+F638 | noteheadLongaDown      |
| U+F639 | noteheadDoubleWholeAlt |
| :      |                        |



```
'withx-black
```

```

:
U+F680    noteheadBlackWithX

```



'parenthesised

|        |                           |
|--------|---------------------------|
| U+F5DF | noteheadDoubleWholeParens |
| U+F5DE | noteheadWholeParens       |
| U+F5DD | noteheadHalfParens        |
| U+F5DC | noteheadBlackParens       |

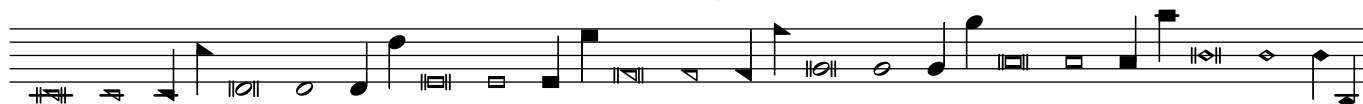


## Shape Note Heads

All forms in LilyPond are supported, but some note heads of Feta don't have exact matches in SMuFL, e.g. the thin shapes of `\southernHarmonyHeads` and the reversed shapes for stem up of `\funkHeads`.

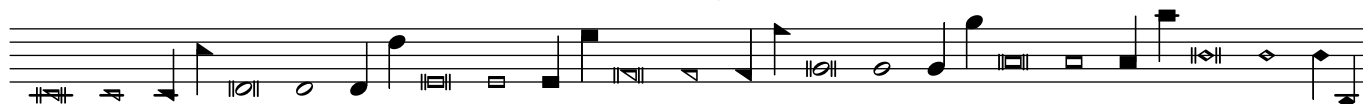
### `\sacredHarpHeads`

|     |        |                                   |
|-----|--------|-----------------------------------|
| fa  | U+ECD3 | noteShapeTriangleLeftDoubleWhole  |
|     | U+E1B6 | noteShapeTriangleLeftWhite        |
|     | U+E1B7 | noteShapeTriangleLeftBlack        |
|     | U+ECD2 | noteShapeTriangleRightDoubleWhole |
|     | U+E1B4 | noteShapeTriangleRightWhite       |
|     | U+E1B5 | noteShapeTriangleRightBlack       |
| sol | U+ECD0 | noteShapeRoundDoubleWhole         |
|     | U+E1B0 | noteShapeRoundWhite               |
|     | U+E1B1 | noteShapeRoundBlack               |
| la  | U+ECD1 | noteShapeSquareDoubleWhole        |
|     | U+E1B2 | noteShapeSquareWhite              |
|     | U+E1B3 | noteShapeSquareBlack              |
| mi  | U+ECD4 | noteShapeDiamondDoubleWhole       |
|     | U+E1B8 | noteShapeDiamondWhite             |
|     | U+E1B9 | noteShapeDiamondBlack             |



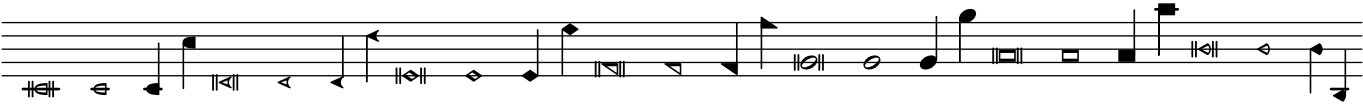
### `\southernHarmonyHeads`

|     |        |                                   |
|-----|--------|-----------------------------------|
| fa  | U+ECD3 | noteShapeTriangleLeftDoubleWhole  |
|     | U+E1B6 | noteShapeTriangleLeftWhite        |
|     | U+E1B7 | noteShapeTriangleLeftBlack        |
|     | U+ECD2 | noteShapeTriangleRightDoubleWhole |
|     | U+E1B4 | noteShapeTriangleRightWhite       |
|     | U+E1B5 | noteShapeTriangleRightBlack       |
| sol | U+ECD0 | noteShapeRoundDoubleWhole         |
|     | U+E1B0 | noteShapeRoundWhite               |
|     | U+E1B1 | noteShapeRoundBlack               |
| la  | U+ECD1 | noteShapeSquareDoubleWhole        |
|     | U+E1B2 | noteShapeSquareWhite              |
|     | U+E1B3 | noteShapeSquareBlack              |
| mi  | U+ECD4 | noteShapeDiamondDoubleWhole       |
|     | U+E1B8 | noteShapeDiamondWhite             |
|     | U+E1B9 | noteShapeDiamondBlack             |



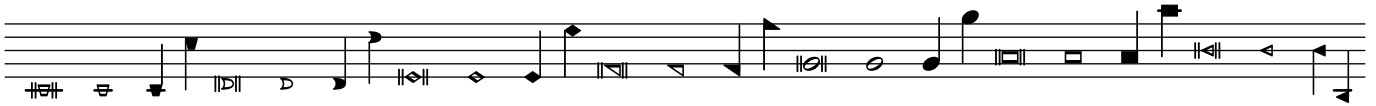
\funkHeads

|     |        |                                       |
|-----|--------|---------------------------------------|
| do  | U+ECDB | noteShapeMoonLeftDoubleWhole          |
|     | U+E1C6 | noteShapeMoonLeftWhite                |
|     | U+E1C7 | noteShapeMoonLeftBlack                |
| re  | U+ECDC | noteShapeArrowheadLeftDoubleWhole     |
|     | U+E1C8 | noteShapeArrowheadLeftWhite           |
|     | U+E1C9 | noteShapeArrowheadLeftBlack           |
| mi  | U+ECD4 | noteShapeDiamondDoubleWhole           |
|     | U+E1B8 | noteShapeDiamondWhite                 |
|     | U+E1B9 | noteShapeDiamondBlack                 |
| fa  | U+ECD3 | noteShapeTriangleLeftDoubleWhole      |
|     | U+E1B6 | noteShapeTriangleLeftWhite            |
|     | U+E1B7 | noteShapeTriangleLeftBlack            |
|     | U+ECD2 | noteShapeTriangleRightDoubleWhole     |
|     | U+E1B4 | noteShapeTriangleRightWhite           |
| sol | U+E1B5 | noteShapeTriangleRightBlack           |
|     | U+ECD0 | noteShapeRoundDoubleWhole             |
|     | U+E1B0 | noteShapeRoundWhite                   |
| la  | U+E1B1 | noteShapeRoundBlack                   |
|     | U+ECD1 | noteShapeSquareDoubleWhole            |
|     | U+E1B2 | noteShapeSquareWhite                  |
| ti  | U+E1B3 | noteShapeSquareBlack                  |
|     | U+ECDD | noteShapeTriangleRoundLeftDoubleWhole |
|     | U+E1CA | noteShapeTriangleRoundLeftWhite       |
|     | U+E1CB | noteShapeTriangleRoundLeftBlack       |



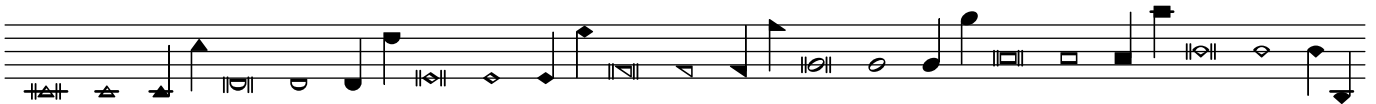
## \walkerHeads

|     |        |                                       |
|-----|--------|---------------------------------------|
| do  | U+ECD8 | noteShapeKeystoneDoubleWhole          |
|     | U+E1C0 | noteShapeKeystoneWhite                |
|     | U+E1C1 | noteShapeKeystoneBlack                |
| re  | U+ECD9 | noteShapeQuarterMoonDoubleWhole       |
|     | U+E1C2 | noteShapeQuarterMoonWhite             |
|     | U+E1C3 | noteShapeQuarterMoonBlack             |
| mi  | U+ECD4 | noteShapeDiamondDoubleWhole           |
|     | U+E1B8 | noteShapeDiamondWhite                 |
|     | U+E1B9 | noteShapeDiamondBlack                 |
| fa  | U+ECD3 | noteShapeTriangleLeftDoubleWhole      |
|     | U+E1B6 | noteShapeTriangleLeftWhite            |
|     | U+E1B7 | noteShapeTriangleLeftBlack            |
|     | U+ECD2 | noteShapeTriangleRightDoubleWhole     |
|     | U+E1B4 | noteShapeTriangleRightWhite           |
| sol | U+E1B5 | noteShapeTriangleRightBlack           |
|     | U+ECD0 | noteShapeRoundDoubleWhole             |
|     | U+E1B0 | noteShapeRoundWhite                   |
| la  | U+E1B1 | noteShapeRoundBlack                   |
|     | U+ECD1 | noteShapeSquareDoubleWhole            |
|     | U+E1B2 | noteShapeSquareWhite                  |
| ti  | U+E1B3 | noteShapeSquareBlack                  |
|     | U+ECDA | noteShapeIsoscelesTriangleDoubleWhole |
|     | U+E1C4 | noteShapeIsoscelesTriangleWhite       |
|     | U+E1C5 | noteShapeIsoscelesTriangleBlack       |



## \aikenHeads

|     |        |                                   |
|-----|--------|-----------------------------------|
| do  | U+ECD5 | noteShapeTriangleUpDoubleWhole    |
|     | U+E1BA | noteShapeTriangleUpWhite          |
|     | U+E1BB | noteShapeTriangleUpBlack          |
| re  | U+ECD6 | noteShapeMoonDoubleWhole          |
|     | U+E1BC | noteShapeMoonWhite                |
|     | U+E1BD | noteShapeMoonBlack                |
| mi  | U+ECD4 | noteShapeDiamondDoubleWhole       |
|     | U+E1B8 | noteShapeDiamondWhite             |
|     | U+E1B9 | noteShapeDiamondBlack             |
| fa  | U+ECD3 | noteShapeTriangleLeftDoubleWhole  |
|     | U+E1B6 | noteShapeTriangleLeftWhite        |
|     | U+E1B7 | noteShapeTriangleLeftBlack        |
|     | U+ECD2 | noteShapeTriangleRightDoubleWhole |
|     | U+E1B4 | noteShapeTriangleRightWhite       |
| sol | U+E1B5 | noteShapeTriangleRightBlack       |
|     | U+ECD0 | noteShapeRoundDoubleWhole         |
|     | U+E1B0 | noteShapeRoundWhite               |
| la  | U+E1B1 | noteShapeRoundBlack               |
|     | U+ECD1 | noteShapeSquareDoubleWhole        |
|     | U+E1B2 | noteShapeSquareWhite              |
| ti  | U+E1B3 | noteShapeSquareBlack              |
|     | U+ECD7 | noteShapeTriangleRoundDoubleWhole |
|     | U+E1BE | noteShapeTriangleRoundWhite       |
|     | U+E1BF | noteShapeTriangleRoundBlack       |

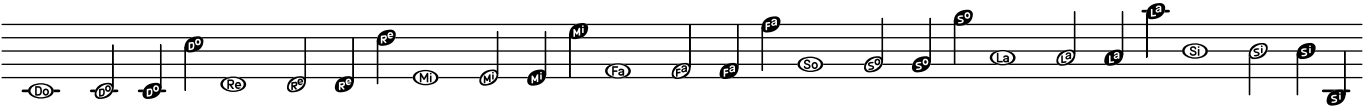


Note Name Note Heads

\ekmNameHeads...

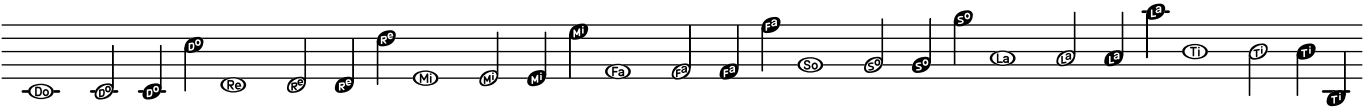
Draw note heads with solfège (easy play) note names. [Err]

|                    |        |             |
|--------------------|--------|-------------|
| \ekmNameHeads      |        |             |
| \ekmNameHeadsMinor |        |             |
| do                 | U+E150 | noteDoWhole |
|                    | U+E158 | noteDoHalf  |
|                    | U+E160 | noteDoBlack |
| re                 | U+E151 | noteReWhole |
|                    | U+E159 | noteReHalf  |
|                    | U+E161 | noteReBlack |
| mi                 | U+E152 | noteMiWhole |
|                    | U+E15A | noteMiHalf  |
|                    | U+E162 | noteMiBlack |
| fa                 | U+E153 | noteFaWhole |
|                    | U+E15B | noteFaHalf  |
|                    | U+E163 | noteFaBlack |
| so                 | U+E154 | noteSoWhole |
|                    | U+E15C | noteSoHalf  |
|                    | U+E164 | noteSoBlack |
| la                 | U+E155 | noteLaWhole |
|                    | U+E15D | noteLaHalf  |
|                    | U+E165 | noteLaBlack |
| si                 | U+E157 | noteSiWhole |
|                    | U+E15F | noteSiHalf  |
|                    | U+E167 | noteSiBlack |



\ekmNameHeadsTi  
\ekmNameHeadsTiMinor

|           |        |             |
|-----------|--------|-------------|
| do ... la | :      |             |
| ti        | U+E156 | noteTiWhole |
|           | U+E15E | noteTiHalf  |
|           | U+E166 | noteTiBlack |



## Note Clusters

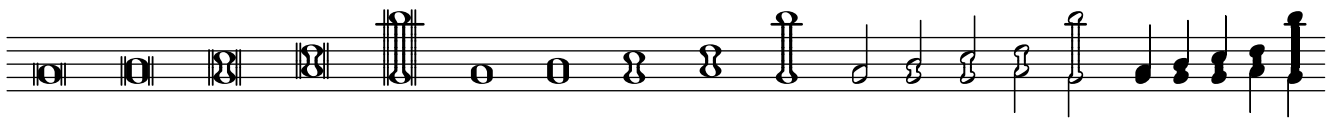
`\ekmMakeClusters MUSIC`

Draw clusters instead of chords in MUSIC, consisting of a bottom and a top note head, and ignoring inner notes of the chords ("Cowell clusters").

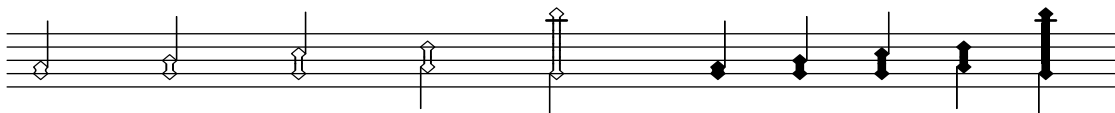
The style can be set with `\override NoteHead.style = #STYLE`

### STYLE

|          |        |                                  |
|----------|--------|----------------------------------|
| 'default | U+E124 | noteheadClusterDoubleWhole2nd    |
|          | U+E128 | noteheadClusterDoubleWhole3rd    |
|          | U+E12C | noteheadClusterDoubleWholeTop    |
|          | U+E12D | noteheadClusterDoubleWholeMiddle |
|          | U+E12E | noteheadClusterDoubleWholeBottom |
|          | U+E125 | noteheadClusterWhole2nd          |
|          | U+E129 | noteheadClusterWhole3rd          |
|          | U+E12F | noteheadClusterWholeTop          |
|          | U+E130 | noteheadClusterWholeMiddle       |
|          | U+E131 | noteheadClusterWholeBottom       |
|          | U+E126 | noteheadClusterHalf2nd           |
|          | U+E12A | noteheadClusterHalf3rd           |
|          | U+E132 | noteheadClusterHalfTop           |
|          | U+E133 | noteheadClusterHalfMiddle        |
|          | U+E134 | noteheadClusterHalfBottom        |
|          | U+E127 | noteheadClusterQuarter2nd        |
|          | U+E12B | noteheadClusterQuarter3rd        |
|          | U+E135 | noteheadClusterQuarterTop        |
|          | U+E136 | noteheadClusterQuarterMiddle     |
|          | U+E137 | noteheadClusterQuarterBottom     |



|           |        |                                   |
|-----------|--------|-----------------------------------|
| 'harmonic | U+E138 | noteheadDiamondClusterWhite2nd    |
|           | U+E13A | noteheadDiamondClusterWhite3rd    |
|           | U+E13C | noteheadDiamondClusterWhiteTop    |
|           | U+E13D | noteheadDiamondClusterWhiteMiddle |
|           | U+E13E | noteheadDiamondClusterWhiteBottom |
|           | U+E139 | noteheadDiamondClusterBlack2nd    |
|           | U+E13B | noteheadDiamondClusterBlack3rd    |
|           | U+E13F | noteheadDiamondClusterBlackTop    |
|           | U+E140 | noteheadDiamondClusterBlackMiddle |
|           | U+E141 | noteheadDiamondClusterBlackBottom |



'square

U+E145

U+E146

U+E147

U+E142

U+E143

U+E144

noteheadRectangularClusterWhiteTop

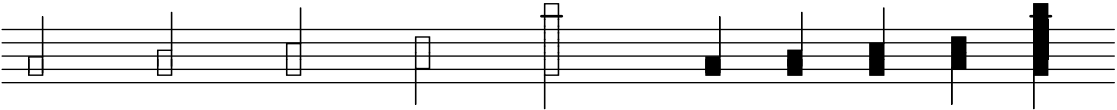
noteheadRectangularClusterWhiteMiddle

noteheadRectangularClusterWhiteBottom

noteheadRectangularClusterBlackTop

noteheadRectangularClusterBlackMiddle

noteheadRectangularClusterBlackBottom



with Ekmelos

'diamond

U+F64B

U+F64C

U+F64D

U+F64E

U+F64F

:

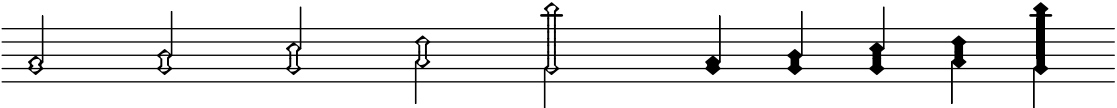
noteheadDiamondClusterHalf2nd

noteheadDiamondClusterHalf3rd

noteheadDiamondClusterHalfTop

noteheadDiamondClusterHalfMiddle

noteheadDiamondClusterHalfBottom



Note: For intervals larger than a third (except for 'square) the drawn cluster is a stack of one bottom segment, M middle segments, and one top segment. Mid and Top are the staff positions of the middle and top segments relative to the bottom segment.

| Interval | M | Mid     | Top |
|----------|---|---------|-----|
| 4th      | 0 | -       | 3   |
| 5th      | 1 | 2       | 4   |
| 6th      | 2 | 2 3     | 5   |
| 7th      | 3 | 2 3 4   | 6   |
| octave   | 4 | 2 3 4 5 | 7   |
| ...      |   |         |     |

The segment glyphs in [Ekmelos](#) are designed for these values.  
However, in the implementation notes of [SMuFL](#) "Note clusters" , the octave cluster is said to have 3 middle segments, while the 6th cluster has 2 middle segments. The “appropriate number of middle segments” varies apparently depending on the font.

## Note Markup

`\ekm-note DURATION DIRECTION`

Draw a note with DURATION and stem DIRECTION as markup.

DIRECTION can be fractional for another stem length or 0 for no stem.

The property `style` can be any style of a [note head](#) or of a precomposed note (see below). In the latter case, no extra stem or flag is drawn, only the sign of DIRECTION is respected, and `flag-style` is ignored.

Used properties: See `\ekm-note-by-number`.

`\ekm-note-by-number LOG DOT-COUNT DIRECTION`

Draw a note with duration LOG, DOT-COUNT augmentation dots, and stem DIRECTION as markup.

LOG is usually in the range -1 to 10.

Used properties:

- `font-size (0)`
- `style ('')`
- `flag-style ('')`
- `dots-direction (0)`

## Precomposed notes

### STYLE

|       |        |                 |
|-------|--------|-----------------|
| 'note | U+E1D0 | noteDoubleWhole |
|       | U+E1D2 | noteWhole       |
|       | :      |                 |
|       | U+E1E6 | note1024thDown  |



|            |        |                    |
|------------|--------|--------------------|
| 'metronome | U+ECA0 | metNoteDoubleWhole |
|            | U+ECA2 | metNoteWhole       |
|            | :      |                    |
|            | U+ECB6 | metNote1024thDown  |



with Ekmelos

'note



'note-short



'note-straight



'note-beamed



\ekm-rhythm DEFINITION

Draw a rhythmic pattern according to **DEFINITION** as markup. Space between tokens is ignored.  
A pattern is composed of SMuFL beamed group notes and stem-up precomposed notes (see above), either the short stem beamed notes and 'metronome style notes, or if DEFINITION has the prefix \* , the long stem beamed notes and 'note style notes.

|         |                                                                                     |        |                                |
|---------|-------------------------------------------------------------------------------------|--------|--------------------------------|
| 1       |    | U+ECA2 | metNoteWhole                   |
| 2       |    | U+ECA3 | metNoteHalfUp                  |
| 4       |    | U+ECA5 | metNoteQuarterUp               |
| 8       |    | U+ECA7 | metNote8thUp                   |
| 16      |    | U+ECA9 | metNote16thUp                  |
| 32      |    | U+ECAB | metNote32ndUp                  |
| n       |    | U+E1F0 | textBlackNoteShortStem         |
| .       |    | U+E1FC | textAugmentationDot            |
| 8 [     |    |        |                                |
| 8 . [   |  |        |                                |
| 8 . . [ |  |        |                                |
| 8 -     |  |        |                                |
| 16 [    |  |        |                                |
| 16 . [  |  |        |                                |
| 16 -    |  |        |                                |
| 16 +    |  |        |                                |
| 32 [    |  |        |                                |
| 32 -    |  |        |                                |
| 32 +    |  |        |                                |
| 8 ]     |  | U+E1F2 | textBlackNoteFrac8thShortStem  |
| 16 ]    |  | U+E1F4 | textBlackNoteFrac16thShortStem |
| 32 ]    |  | U+E1F6 | textBlackNoteFrac32ndLongStem  |
| 8 _     |  | U+E1F7 | textCont8thBeamShortStem       |
| 16 _    |  | U+E1F9 | textCont16thBeamShortStem      |
| 32 _    |  | U+E1FB | textCont32ndBeamLongStem       |

|   |   |        |                                 |
|---|---|--------|---------------------------------|
| ( | ⌈ | U+E1FE | textTupletBracketStartShortStem |
| 3 | 3 | U+E1FF | textTuplet3ShortStem            |
| ) | ⌋ | U+E200 | textTupletBracketEndShortStem   |
| ~ | ~ | U+E1FD | textTie                         |
| = | = | U+E08F | timeSigEquals                   |

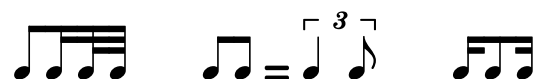
= draws the equal sign with space around.

(3 draws the two glyphs with a space in between.

## Examples

```
\ekm-rhythm "*8[ 16+ 32+ 32]"
```

```
\ekm-rhythm "8[ 8] = n (3 8 )"
\ekm-rhythm "16[ 8- 16]"
```



## Augmentation Dots

---

\ekmSmuflOn #'dot

Draw SMuFL augmentation dots.

.                      .                      U+E1E7      augmentationDot

## Flags and Grace Note Slashes

---

`\ekmSmuflOn #'flag`

Draw SMuFL flags and grace note slashes with `\slashedGrace`.

The style can be set with `\override Flag.style = #STYLE`

### STYLE

`'default`

`U+E240 flag8thUp`

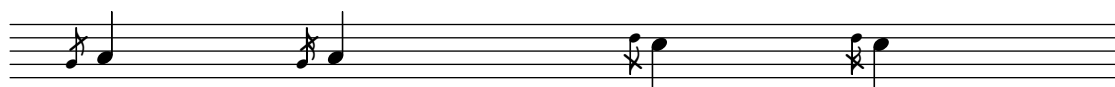
`:`

`U+E24F flag1024thDown`



`U+E564 graceNoteSlashStemUp`

`U+E565 graceNoteSlashStemDown`



with Bravura or Ekmelos

'short



'straight



## Rests

---

```
\ekmSmuflOn #'rest
```

Draw SMuFL rests and multi-measure rests, as well as SMuFL time signature digits for multi-measure rest numbers.

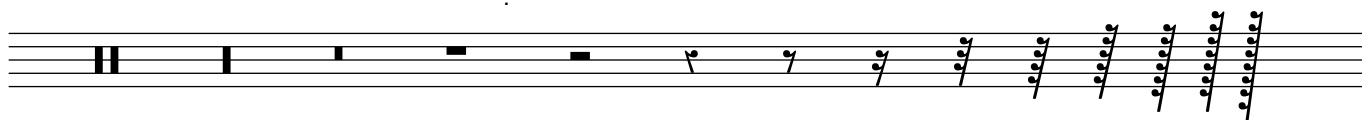
The style can be set with `\override Rest.style = #STYLE`

### STYLE

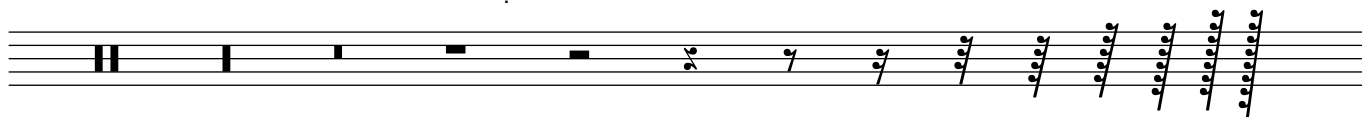
```
'default          U+E4E0  restMaxima
                  :
                  U+E4ED  rest1024th
```



```
'classical        :
                  U+E4F2  restQuarterOld
                  :
```



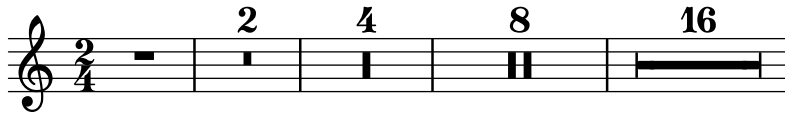
```
'z                :
                  U+E4F6  restQuarterZ
                  :
```



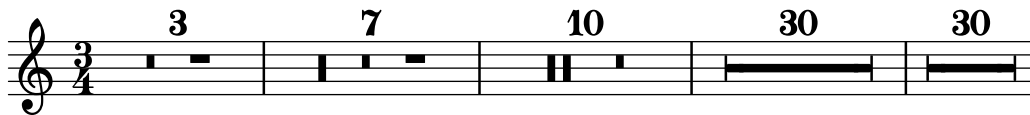
## Examples

Here, the multi-measure rest numbers are SMuFL glyphs, while the time signatures are from LilyPond's Emmentaler font.

```
\relative c'' {
  \ekmSmuflOn #'rest
  \compressMMRests {
    \time 2/4
    R2 R1 R\breve R\longa R\maxima
  }
}
```



```
\relative c'' {
  \ekmSmuflOn #'rest
  \compressMMRests {
    \time 3/4
    R2.*3
    R2.*7
    R2.*10
    R2.*30
    \override MultiMeasureRest.space-increment = -2.5
    R2.*30
  }
}
```



## Rest Markup

---

`\ekm-rest DURATION`

Draw either a rest or a multi-measure rest with DURATION as markup.

`\ekm-rest-by-number LOG DOT-COUNT`

Draw a rest with duration LOG and DOT-COUNT augmentation dots as markup. The dots are vertically centered, contrary to `\rest-by-number`.

LOG is in the range -3 to 10.

Used properties:

- `font-size` (0)
- `ledgers` '(-1 0 1))
- `style` '())

`\ekm-multi-measure-rest-by-number MEASURES`

Draw a multi-measure rest as markup, with the number placed centered above unless it is 1.

Used properties:

- `font-size` (0)
- `expand-limit` (10)
- `style` '())
- `word-space`
- `width` (8)
- `multi-measure-rest-number` (#t)

## Examples

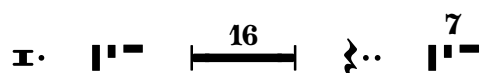
```
\ekm-rest { \breve. }
```

```
\override #'(multi-measure-rest . #t)
\override #'(multi-measure-rest-number . #f)
\ekm-rest { 1*7 }
```

```
\override #'(multi-measure-rest . #t)
\ekm-rest { 1*16 }
```

```
\ekm-rest-by-number #2 #2
```

```
\ekm-multi-measure-rest-by-number #7
```



## Scripts and Expressive Marks

`\ekmSmuflOn #'script`

Draw SMuFL scripts for expressive marks like articulations, ornamentations, performance indications, fermatas, repeat signs, etc.

`\ekmScript SCRIPT-NAME #'(EXTEXT-UP . EXTEXT-DOWN)`

`\ekmScript SCRIPT-NAME EXTEXT`

Create a script from [EXTEXT](#), either a pair for up and down, or a single value for both directions. If the latter is a list it must be enclosed in a list. [\[LP\]](#)

SCRIPT-NAME is the symbol of an existing script like `accent`, `marcato`, `trill`, `turn`, `upbow`, `open`, `segno`, etc. It determines the vertical positioning of the script.

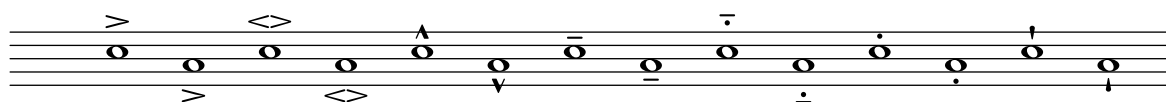
`\ekmScriptSmall SCRIPT-NAME #'(EXTEXT-UP . EXTEXT-DOWN)`

`\ekmScriptSmall SCRIPT-NAME EXTEXT`

Create a script with a 3 steps smaller font size. [\[LP\]](#)

### Articulations

|                             |        |                                       |
|-----------------------------|--------|---------------------------------------|
| <code>\accent</code>        | U+E4A0 | <code>articAccentAbove</code>         |
|                             | U+E4A1 | <code>articAccentBelow</code>         |
| <code>\espressivo</code>    | U+ED40 | <code>articSoftAccentAbove</code>     |
|                             | U+ED41 | <code>articSoftAccentBelow</code>     |
| <code>\marcato</code>       | U+E4AC | <code>articMarcatoAbove</code>        |
|                             | U+E4AD | <code>articMarcatoBelow</code>        |
| <code>\tenuto</code>        | U+E4A4 | <code>articTenutoAbove</code>         |
|                             | U+E4A5 | <code>articTenutoBelow</code>         |
| <code>\portato</code>       | U+E4B2 | <code>articTenutoStaccatoAbove</code> |
|                             | U+E4B3 | <code>articTenutoStaccatoBelow</code> |
| <code>\staccato</code>      | U+E4A2 | <code>articStaccatoAbove</code>       |
|                             | U+E4A3 | <code>articStaccatoBelow</code>       |
| <code>\staccatissimo</code> | U+E4A6 | <code>articStaccatissimoAbove</code>  |
|                             | U+E4A7 | <code>articStaccatissimoBelow</code>  |



### Ornamentations

|                           |        |                                   |
|---------------------------|--------|-----------------------------------|
| <code>\trill</code>       | U+E566 | <code>ornamentTrill</code>        |
| <code>\turn</code>        | U+E567 | <code>ornamentTurn</code>         |
| <code>\reverseturn</code> | U+E568 | <code>ornamentTurnInverted</code> |
| <code>\slashturn</code>   | U+E569 | <code>ornamentTurnSlash</code>    |
| <code>\haydnturn</code>   | U+E56F | <code>ornamentHaydn</code>        |

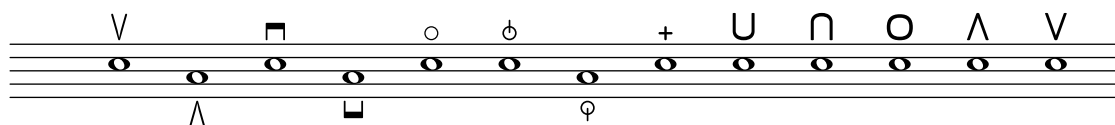


|                            |     |        |                                           |
|----------------------------|-----|--------|-------------------------------------------|
| <code>\prall</code>        |     | U+E56C | ornamentShortTrill                        |
| <code>\prallprall</code>   |     | U+E56E | ornamentTremblement                       |
| <code>\mordent</code>      |     | U+E56D | ornamentMordent                           |
| <code>\prallmordent</code> |     | U+E5BD | ornamentPrecompTrillWithMordent           |
| <code>\upprall</code>      | 2 × | U+E59A | ornamentBottomLeftConcaveStroke           |
|                            |     | U+E59D | ornamentZigZagLineNoRightEnd              |
|                            |     | U+E59E | ornamentZigZagLineWithRightEnd            |
| <code>\downprall</code>    |     | U+E5C6 | ornamentPrecompMordentUpperPrefix         |
| <code>\upmordent</code>    |     | U+E5B8 | ornamentPrecompSlideTrillBach             |
| <code>\downmordent</code>  |     | U+E5C7 | ornamentPrecompInvertedMordentUpperPrefix |
| <code>\prallup</code>      | 3 × | U+E59D | ornamentZigZagLineNoRightEnd              |
|                            |     | U+E5A4 | ornamentRightVerticalStroke               |
| <code>\pralldown</code>    |     | U+E5C8 | ornamentPrecompTrillLowerSuffix           |
| <code>\lineprall</code>    |     | U+E5B2 | ornamentPrecompAppoggTrill                |



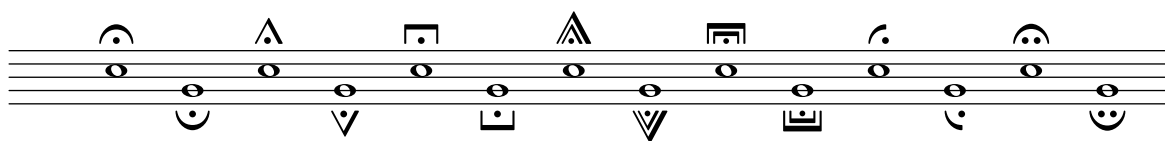
### Performance indications

|                             |        |                           |
|-----------------------------|--------|---------------------------|
| <code>\upbow</code>         | U+E612 | stringsUpBow              |
|                             | U+E613 | stringsUpBowTurned        |
| <code>\downbow</code>       | U+E610 | stringsDownBow            |
|                             | U+E611 | stringsDownBowTurned      |
| <code>\flageolet</code>     | U+E614 | stringsHarmonic           |
| <code>\snappizzicato</code> | U+E631 | pluckedSnapPizzicatoAbove |
|                             | U+E630 | pluckedSnapPizzicatoBelow |
| <code>\stopped</code>       | U+E633 | pluckedLeftHandPizzicato  |
| <code>\heel</code>          | U+E661 | keyboardPedalHeel1        |
| <code>\lheel</code>         |        |                           |
| <code>\varheel</code>       | U+E662 | keyboardPedalHeel2        |
| <code>\rheel</code>         |        |                           |
| <code>\heelcircle</code>    | U+E663 | keyboardPedalHeel3        |
| <code>\toe</code>           | U+E665 | keyboardPedalToe2         |
| <code>\rtoe</code>          |        |                           |
| <code>\vartoe</code>        | U+E664 | keyboardPedalToe1         |
| <code>\lttoe</code>         |        |                           |



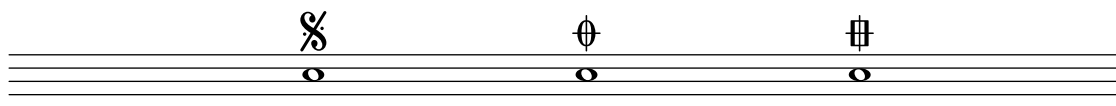
## Fermatas

|                                 |        |                        |
|---------------------------------|--------|------------------------|
| <code>\fermata</code>           | U+E4C0 | fermataAbove           |
|                                 | U+E4C1 | fermataBelow           |
| <code>\shortfermata</code>      | U+E4C4 | fermataShortAbove      |
|                                 | U+E4C5 | fermataShortBelow      |
| <code>\longfermata</code>       | U+E4C6 | fermataLongAbove       |
|                                 | U+E4C7 | fermataLongBelow       |
| <code>\veryshortfermata</code>  | U+E4C2 | fermataVeryShortAbove  |
|                                 | U+E4C3 | fermataVeryShortBelow  |
| <code>\verylongfermata</code>   | U+E4C8 | fermataVeryLongAbove   |
|                                 | U+E4C9 | fermataVeryLongBelow   |
| <code>\henzeshortfermata</code> | U+E4CC | fermataShortHenzeAbove |
|                                 | U+E4CD | fermataShortHenzeBelow |
| <code>\henzelongfermata</code>  | U+E4CA | fermataLongHenzeAbove  |
|                                 | U+E4CB | fermataLongHenzeBelow  |



## Repeat signs

|                       |        |            |
|-----------------------|--------|------------|
| <code>\segno</code>   | U+E047 | segno      |
| <code>\coda</code>    | U+E048 | coda       |
| <code>\varcoda</code> | U+E049 | codaSquare |



## Examples

```

marcatoTenuto =
  \ekmScript #'marcato #'(#xE4BC . #xE4BD)

stringsChangeBowDirection =
  \ekmScript #'downbow #'((#xE626 1))

\relative c'' {
  \ekmSmuflOn #'script

  c4 ^ \marcatoTenuto
  a4 _ \marcatoTenuto

  c4 ^ \ekmScript #'turn ##xE56A

  c4 ^ \ekmScript #'upbow ##xE61C

  c4 ^ \stringsChangeBowDirection

  % use Ekmelos glyphs
  c4 ^ \open
  c4 ^ \halfopen
  c4 ^ #(make-articulation 'halfopenvertical)
}

```



## Dynamics

\ekmSmuflOn #'dynamic

Draw SMuFL absolute dynamic marks, using \ekm-dynamic .

\ekm-dynamic DEFINITION

Draw a dynamic symbol according to [DEFINITION](#) as markup.

DEFINITION must be either a single token or a sequence of the letters f, m, n, p, r, s, z, whose corresponding symbols are concatenated. This is slightly different from the usual interpretation of definition strings.

|        |                      |        |                            |
|--------|----------------------|--------|----------------------------|
| p      | <b><i>p</i></b>      | U+E520 | dynamicPiano               |
| m      | <b><i>m</i></b>      | U+E521 | dynamicMezzo               |
| f      | <b><i>f</i></b>      | U+E522 | dynamicForte               |
| r      | <b><i>r</i></b>      | U+E523 | dynamicRinforzando         |
| s      | <b><i>s</i></b>      | U+E524 | dynamicSforzando           |
| z      | <b><i>z</i></b>      | U+E525 | dynamicZ                   |
| n      | <b><i>n</i></b>      | U+E526 | dynamicNiente              |
| mp     | <b><i>mp</i></b>     | U+E52C | dynamicMP                  |
| mf     | <b><i>mf</i></b>     | U+E52D | dynamicMF                  |
| pf     | <b><i>pf</i></b>     | U+E52E | dynamicPF                  |
| fp     | <b><i>fp</i></b>     | U+E534 | dynamicFortePiano          |
| pppppp | <b><i>pppppp</i></b> | U+E527 | dynamicPPPPPP              |
|        | :                    |        |                            |
| pp     | <b><i>pp</i></b>     | U+E52B | dynamicPP                  |
| ff     | <b><i>ff</i></b>     | U+E52F | dynamicFF                  |
|        | :                    |        |                            |
| ffffff | <b><i>ffffff</i></b> | U+E533 | dynamicFFFFFF              |
| fz     | <b><i>fz</i></b>     | U+E535 | dynamicForzando            |
| sf     | <b><i>sf</i></b>     | U+E536 | dynamicSforzando1          |
| sfp    | <b><i>sfp</i></b>    | U+E537 | dynamicSforzandoPiano      |
| sftp   | <b><i>sftp</i></b>   | U+E538 | dynamicSforzandoPianissimo |
| sfz    | <b><i>sfz</i></b>    | U+E539 | dynamicSforzato            |
| sfzp   | <b><i>sfzp</i></b>   | U+E53A | dynamicSforzatoPiano       |
| sffz   | <b><i>sffz</i></b>   | U+E53B | dynamicSforzatoFF          |
| rf     | <b><i>rf</i></b>     | U+E53C | dynamicRinforzando1        |
| rfz    | <b><i>rfz</i></b>    | U+E53D | dynamicRinforzando2        |

with Ekmelos

|        |                      |        |                     |
|--------|----------------------|--------|---------------------|
| sff    | <b><i>sff</i></b>    | U+F645 | dynamicSforzandoFF  |
| sp     | <b><i>sp</i></b>     | U+F646 | dynamicSP           |
| spp    | <b><i>spp</i></b>    | U+F647 | dynamicSPP          |
| sfffz  | <b><i>sfffz</i></b>  | U+F6F4 | dynamicSforzatoFFF  |
| sffffz | <b><i>sffffz</i></b> | U+F6F5 | dynamicSforzatoFFFF |

\ekmParensDyn STYLE DYNAMIC-MARK

Draw DYNAMIC-MARK (an event) parenthesized with the normal text font.

STYLE

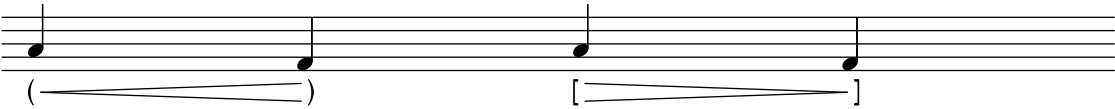
|          |     |
|----------|-----|
| 'default | ( ) |
| 'bracket | [ ] |
| 'brace   | { } |
| 'angle   | < > |

\ekmParensHairpin STYLE

Draw the subsequent hairpin parenthesized.

STYLE

|          |        |                                |
|----------|--------|--------------------------------|
| 'default | U+E542 | dynamicHairpinParenthesisLeft  |
|          | U+E543 | dynamicHairpinParenthesisRight |
| 'bracket | U+E544 | dynamicHairpinBracketLeft      |
|          | U+E545 | dynamicHairpinBracketRight     |



## Laissez Vibrer

---

`\ekmSmuflOn #'lv`

Draw SMuFL laissez vibrer ties.

`\ekmLaissezVibrer SIZE`

Draw a laissez vibrer tie after a note, corresponding to SIZE. [\[LP\]](#)

`\laissezVibrer` is equivalent to `\ekmLaissezVibrer #0`

### SIZE

|            |   |        |                         |
|------------|---|--------|-------------------------|
| any number | ⌒ | U+E4BA | articLaissezVibrerAbove |
|            | ⏟ | U+E4BB | articLaissezVibrerBelow |

### with Ekmelos

|       |   |        |                                  |
|-------|---|--------|----------------------------------|
| ≤ 5.5 | ⌒ | U+E4BA | articLaissezVibrerAbove          |
|       | ⏟ | U+E4BB | articLaissezVibrerBelow          |
| ≤ 8   | ⌒ | U+F6FC | articLaissezVibrerAboveLong      |
|       | ⏟ | U+F6FD | articLaissezVibrerBelowLong      |
| > 8   | ⌒ | U+F6FE | articLaissezVibrerAboveExtraLong |
|       | ⏟ | U+F6FF | articLaissezVibrerBelowExtraLong |

## Examples

```
\relative c' {
  \ekmSmuflOn #'lv

  <c f g>1 \ekmLaissezVibrer #0

  % use Ekmelos glyphs
  <e g c>1 \ekmLaissezVibrer #9
}
```



# Tremolo Marks

`\ekmSmuflOn #'tremolo`

Draw SMuFL tremolo marks on stems, except for beamed notes.  
For cue notes, the tremolo marks are taken from style `'cue` if provided by the selected font. Else they are composed using the symbol with 1 flag of the current style in a compressed shape.

The style can be set with `\override StemTremolo.shape = #STYLE`

## STYLE

|            |                                                                                   |        |                  |  |
|------------|-----------------------------------------------------------------------------------|--------|------------------|--|
| 'beam-like |                                                                                   |        |                  |  |
| :8         |  | U+E220 | tremolo1         |  |
|            | :                                                                                 |        |                  |  |
| :128       |  | U+E224 | tremolo5         |  |
| 'fingered  |                                                                                   |        |                  |  |
| :8         |  | U+E225 | tremoloFingered1 |  |
|            | :                                                                                 |        |                  |  |
| :128       |  | U+E229 | tremoloFingered5 |  |

`\ekmTremolo EXTEXT MUSIC`

Draw a tremolo mark from **EXTEXT** on the stems of the tremolo notes in **MUSIC**, independent of the subdivision `:N`, and independent of the `tremolo` switch. A list of code points or a markup is centered horizontally, while a single code point is assumed being a centered stem decoration.  
Some strings are defined as names of symbols:

|               |                                                                                     |        |                         |  |
|---------------|-------------------------------------------------------------------------------------|--------|-------------------------|--|
| "buzzroll"    |  | U+E22A | buzzRoll                |  |
| "penderecki"  |  | U+E22B | pendereckiTremolo       |  |
| "stockhausen" |  | U+E232 | stockhausenTremolo      |  |
| "unmeasured"  |  | U+E22C | unmeasuredTremolo       |  |
| "unmeasuredS" |  | U+E22D | unmeasuredTremoloSimple |  |

## Examples

```

\new Staff \with {
  \ekmSmuflOn #'tremolo
}
\relative c' {
  d1:32
  g4:32 b:16
  c8:16 b:16
  \ekmTremolo stockhausen g4:16
  <<
  {
    \override MultiMeasureRest.staff-position = #2
    R1*2
  }
  \new CueVoice \relative c' {
    d1:32
    g4:32 b:16
    c8:16 b:16
    \ekmTremolo stockhausen g4:16
  }
  >>
}

```



## Stem Decorations

---

\ekmStem EXTEXT MUSIC

Draw a symbol from [EXTEXT](#) vertically centered on the stems in MUSIC. A list of code points or a markup is centered horizontally, while a single code point is assumed being a centered stem decoration. Some strings are defined as names of symbols:

|                   |   |        |                           |
|-------------------|---|--------|---------------------------|
| "sprechgesang"    | × | U+E645 | vocalSprechgesang         |
| "halbGesungen"    | ˘ | U+E64B | vocalHalbGesungen         |
| "sussurando"      | ſ | U+E646 | vocalsSussurando          |
| "damp"            | ⊖ | U+E63B | pluckedDampOnStem         |
| "stringNoise"     | ⚡ | U+E694 | harpStringNoiseStem       |
| "multiphonics"    | 𐀡 | U+E607 | windMultiphonicsBlackStem |
| "bowBehindBridge" | ↵ | U+E618 | stringsBowBehindBridge    |
| "bowOnBridge"     | ⤿ | U+E619 | stringsBowOnBridge        |
| "bowOnTailpiece"  | └ | U+E61A | stringsBowOnTailpiece     |
| "fouette"         | ㄥ | U+E622 | stringsFouette            |
| "vibrato"         | ㄣ | U+E623 | stringsVibratoPulse       |
| "rimShot"         | × | U+E7FD | pictRimShotOnStem         |
| "crush"           | 𐀢 | U+E80C | pictCrushStem             |
| "deadNote"        | × | U+E80D | pictDeadNoteStem          |
| "swish"           | ↗ | U+E808 | pictSwishStem             |
| "turnRight"       | ↪ | U+E809 | pictTurnRightStem         |
| "turnLeft"        | ↩ | U+E80A | pictTurnLeftStem          |
| "turnRightLeft"   | ↻ | U+E80B | pictTurnRightLeftStem     |

## Examples

```

\relative c'' {
  \ekmTremolo ##xE233
  { b4:8 a: }

  % uses a list to center horizontally
  \ekmTremolo #'(xE56C)
  { b4:8 a: }

  \ekmStem sussurando
  { b4 a }

  \ekmStem "S"
  { b a }

  \ekmStem #'(xE843 xE844)
  { b a }

  \ekmStem \markup { \fontsize #-6 \sans "pizz" }
  { b a }
}

```



## Falls and Doits

\ekmBendAfter STYLE DIRECTION

Draw a fall or doit (lift) symbol after a note. Only the sign of DIRECTION is respected, contrary to \bendAfter.

### STYLE

'bend

UP

U+E5D6 brassDoitLong

U+E5D5 brassDoitMedium

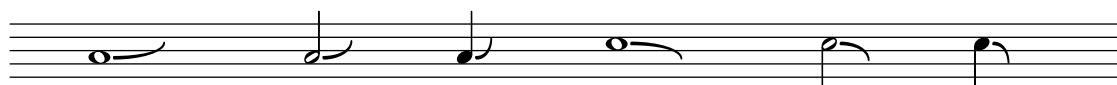
U+E5D4 brassDoitShort

DOWN

U+E5D9 brassFallLipLong

U+E5D8 brassFallLipMedium

U+E5D7 brassFallLipShort



'rough

UP

U+E5D3 brassLiftLong

U+E5D2 brassLiftMedium

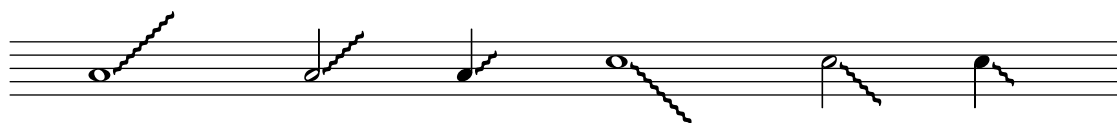
U+E5D1 brassLiftShort

DOWN

U+E5DF brassFallRoughLong

U+E5DE brassFallRoughMedium

U+E5DD brassFallRoughShort



'smooth

UP

U+E5EE brassLiftSmoothLong

U+E5ED brassLiftSmoothMedium

U+E5EC brassLiftSmoothShort

DOWN

U+E5DC brassFallSmoothLong

U+E5DB brassFallSmoothMedium

U+E5DA brassFallSmoothShort



\ekmScoop DIRECTION MUSIC

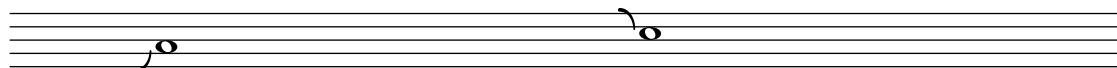
Draw a scoop or plop symbol to the left of each note in MUSIC.

UP

U+E5D0 brassScoop

DOWN

U+E5E0 brassPlop



## Arpeggios

---

`\ekmSmuflOn #'arpeggio`

Draw SMuFL arpeggios.

The style can be set with `\override Arpeggio.style = #STYLE`

### STYLE

|          |                                                                                   |        |                           |
|----------|-----------------------------------------------------------------------------------|--------|---------------------------|
| 'default |  | U+EAA9 | wiggleArpeggiatoUp        |
|          |  | U+EAAA | wiggleArpeggiatoDown      |
|          |  | U+EAAD | wiggleArpeggiatoUpArrow   |
|          |  | U+EAAE | wiggleArpeggiatoDownArrow |
| 'swash   | :                                                                                 |        |                           |
|          |  | U+EAAB | wiggleArpeggiatoUpSwash   |
|          |  | U+EAAC | wiggleArpeggiatoDownSwash |

**Note:** The commands `\ekmArpeggioArrowUp`, `\ekmArpeggioArrowDown`, and `\ekmArpeggioNormal` are deprecated. The corresponding LilyPond commands `\arpeggioArrowUp` etc. now support SMuFL output.

## Examples

```

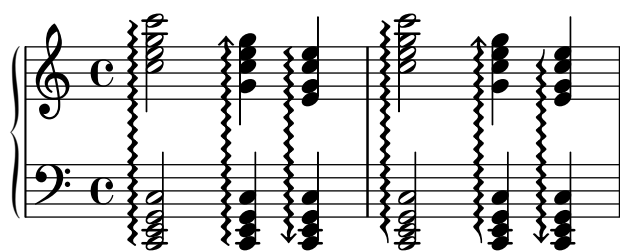
\new PianoStaff \with {
  \ekmSmuflOn #'arpeggio
}
<<
  \set PianoStaff.connectArpeggios = ##t

  \new Staff \relative c'' {
    <c e g c>2 \arpeggio
    \once \override PianoStaff.Arpeggio.arpeggio-direction = #UP
    <g c e g>4 \arpeggio
    \once \override PianoStaff.Arpeggio.arpeggio-direction = #DOWN
    <e g c e>4 \arpeggio

    \override PianoStaff.Arpeggio.style = #'swash
    <c' e g c>2 \arpeggio
    \once \override PianoStaff.Arpeggio.arpeggio-direction = #UP
    <g c e g>4 \arpeggio
    \once \override PianoStaff.Arpeggio.arpeggio-direction = #DOWN
    <e g c e>4 \arpeggio
  }
  \new Staff \relative c, {
    \clef bass
    <c e g c>2 \arpeggio
    <c e g c>4 \arpeggio
    <c e g c>4 \arpeggio

    <c e g c>2 \arpeggio
    <c e g c>4 \arpeggio
    <c e g c>4 \arpeggio
  }
>>

```



## Breathing Signs and Caesuras

---

`\ekmBreathing EXTEXT`

Draw a breathing sign or caesura from [EXTEXT](#) .

### Examples

```
\relative c'' {
  c4
  \ekmBreathing ##xE4CE
  c
  \ekmBreathing ##xE4D1
  c
  \ekmBreathing #'(##xE4D1 1)
  c
}
```



## Percent Repeats

---

`\ekmSmuflOn #'percent`

Draw SMuFL percent repeats.

|                                                                                   |        |                |
|-----------------------------------------------------------------------------------|--------|----------------|
|  | U+E504 | repeatBarSlash |
|  | U+E500 | repeat1Bar     |
|  | U+E501 | repeat2Bars    |

## Examples

```
\relative c'' {
  \ekmSmuflOn #'percent

  \repeat percent 2 { c2 }
  \repeat percent 4 { c4 }
  \repeat percent 4 { c8 d }
  \repeat percent 4 { c16 d e f }
  \repeat percent 5 { c32 d e f }
  \repeat percent 4 { c64 d e f }
  \repeat percent 4 { c128 d e f }
  \break

  \repeat percent 2 { c4 d e d }
  \repeat percent 2 { c2 e }
  \break

  \repeat percent 2 { g,16 b c8~ c4 }
  \repeat percent 2 { c4 d e d | c2 e }
}
```



## Multi-Segment Spanner

---

`\ekmSmuflOn #'textspan`

Draw text spans assembled from SMuFL multi-segment glyphs.

`\ekmStartSpan STYLE TEMPO ATTACHMENT`

Start a text span or [trill span](#). [LP]

STYLE 'trill starts a trill span. A second style can be added to draw other glyphs, e.g.

'trill-vibrato-large. Any other style starts a text span. An undefined style like 'dashed-line produces normal LilyPond output.

TEMPO is a number or a pair of numbers (rounded to integer) for the segments of the spanner. 0 is the main (medium) segment. Positive values mean faster (narrower) segments. Negative values mean slower (wider) segments. A pair ' (A . B) draws all segments from A through B, evenly distributed over the spanner. The available range of numbers depends on the style.

ATTACHMENT is an [EXTEXT](#) or a pair ' (EXTEXT-LEFT . EXTEXT-RIGHT) for the edge symbols. A single [EXTEXT](#) is equivalent to ' (EXTEXT . 0). It must be specified in a pair if itself is a list.

#f draws the standard glyph on the left and right edge according to the style.

`\ekmStartSpanMusic STYLE TEMPO ATTACHMENT MUSIC`

Start a text span or trill span at MUSIC.

This is a music function that doesn't need the `textspan` or `trill` [SMuFL switch](#) turned on.

### STYLE

'trill

|      |                                                                                     |        |                    |
|------|-------------------------------------------------------------------------------------|--------|--------------------|
| left |  | U+E566 | ornamentTrill      |
| 4    |  | U+EAA0 | wiggleTrillFastest |
|      | :                                                                                   |        |                    |
| 0    |  | U+EAA4 | wiggleTrill        |
|      | :                                                                                   |        |                    |
| -4   |  | U+EAA8 | wiggleTrillSlowest |

'vibrato

|      |                                                                                     |        |                            |
|------|-------------------------------------------------------------------------------------|--------|----------------------------|
| left |  | U+EACC | wiggleVibratoStart         |
| 3    |  | U+EADB | wiggleVibratoMediumFastest |
|      | :                                                                                   |        |                            |
| 0    |  | U+EADE | wiggleVibratoMediumFast    |
|      | :                                                                                   |        |                            |
| -3   |  | U+EAE1 | wiggleVibratoMediumSlowest |

'vibrato-small

|      |                                                                                     |        |                           |
|------|-------------------------------------------------------------------------------------|--------|---------------------------|
| left |  | U+EACC | wiggleVibratoStart        |
| 3    |  | U+EAD4 | wiggleVibratoSmallFastest |
|      | :                                                                                   |        |                           |
| 0    |  | U+EAD7 | wiggleVibratoSmallFast    |
|      | :                                                                                   |        |                           |
| -3   |  | U+EADA | wiggleVibratoSmallSlowest |

## 'vibrato-smallest

|      |                                                                                   |        |                              |
|------|-----------------------------------------------------------------------------------|--------|------------------------------|
| left |  | U+EACC | wiggleVibratoStart           |
| 3    |  | U+EACD | wiggleVibratoSmallestFastest |
|      | :                                                                                 |        |                              |
| 0    |  | U+EAD0 | wiggleVibratoSmallestFast    |
|      | :                                                                                 |        |                              |
| -3   |  | U+EAD3 | wiggleVibratoSmallestSlowest |

## 'vibrato-large

|      |                                                                                   |        |                           |
|------|-----------------------------------------------------------------------------------|--------|---------------------------|
| left |  | U+EACC | wiggleVibratoStart        |
| 3    |  | U+EAE2 | wiggleVibratoLargeFastest |
|      | :                                                                                 |        |                           |
| 0    |  | U+EAE5 | wiggleVibratoLargeFast    |
|      | :                                                                                 |        |                           |
| -3   |  | U+EAE8 | wiggleVibratoLargeSlowest |

## 'vibrato-largest

|      |                                                                                     |        |                             |
|------|-------------------------------------------------------------------------------------|--------|-----------------------------|
| left |    | U+EACC | wiggleVibratoStart          |
| 3    |    | U+EAE9 | wiggleVibratoLargestFastest |
|      | :                                                                                   |        |                             |
| 0    |    | U+EAEC | wiggleVibratoLargestFast    |
|      | :                                                                                   |        |                             |
| -3   |  | U+EAEF | wiggleVibratoLargestSlowest |

## 'circular

|       |                                                                                     |        |                       |
|-------|-------------------------------------------------------------------------------------|--------|-----------------------|
| left  |  | U+EAC4 | wiggleCircularStart   |
| 4     |  | U+EACA | wiggleCircularSmall   |
|       | :                                                                                   |        |                       |
| 0     |  | U+EAC9 | wiggleCircular        |
|       | :                                                                                   |        |                       |
| -4    |  | U+EAC5 | wiggleCircularLargest |
| right |  | U+EACB | wiggleCircularEnd     |

## 'circular-constant

|    |                                                                                     |        |                                    |
|----|-------------------------------------------------------------------------------------|--------|------------------------------------|
| 2  |  | U+EAC2 | wiggleCircularConstantLarge        |
| 1  |  | U+EAC0 | wiggleCircularConstant             |
|    | :                                                                                   |        |                                    |
| 0  |  | U+EAC0 | wiggleCircularConstant             |
|    | :                                                                                   |        |                                    |
| -1 |  | U+EAC1 | wiggleCircularConstantFlipped      |
| -2 |  | U+EAC3 | wiggleCircularConstantFlippedLarge |

## 'wavy

|    |                                                                                     |        |                  |
|----|-------------------------------------------------------------------------------------|--------|------------------|
| 1  |  | U+EAB4 | wiggleWavyNarrow |
| 0  |  | U+EAB5 | wiggleWavy       |
| -1 |  | U+EAB6 | wiggleWavyWide   |

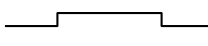
|           |                                                                                   |        |                        |
|-----------|-----------------------------------------------------------------------------------|--------|------------------------|
| 'square   |                                                                                   |        |                        |
| 1         |  | U+EAB7 | wiggleSquareWaveNarrow |
| 0         |  | U+EAB8 | wiggleSquareWave       |
| -1        |  | U+EAB9 | wiggleSquareWaveWide   |
| 'sawtooth |                                                                                   |        |                        |
| 1         |  | U+EABA | wiggleSawtoothNarrow   |
| 0         |  | U+EABB | wiggleSawtooth         |
| -1        |  | U+EABC | wiggleSawtoothWide     |
| 'beam     |                                                                                   |        |                        |
| 7         |  | U+EB02 | beamAccelRit15         |
|           | :                                                                                 |        |                        |
| 0         |  | U+EAFB | beamAccelRit8          |
|           | :                                                                                 |        |                        |
| -7        |  | U+EAF4 | beamAccelRit1          |
| right     |  | U+EB03 | beamAccelRitFinal      |

## with Ekmelos

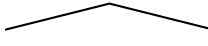
## 'wavy

|        |                                                                                   |        |                         |
|--------|-----------------------------------------------------------------------------------|--------|-------------------------|
| 6 .. 4 |  | U+F7A1 | wiggleWavyExtraNarrow   |
| 3      |  | U+F7A0 | wiggleWavyVeryNarrow    |
| 2      |  | U+F6B3 | wiggleWavyNarrower      |
| 1      |  | U+EAB4 | wiggleWavyNarrow        |
| 0      |  | U+EAB5 | wiggleWavy              |
| -1     |  | U+EAB6 | wiggleWavyWide          |
| -2     |  | U+F6B4 | wiggleWavyWider         |
| :      |                                                                                   |        |                         |
| -6     |  | U+F727 | wiggleWavyQuadrupleWide |

## 'square

|        |                                                                                    |        |                               |
|--------|------------------------------------------------------------------------------------|--------|-------------------------------|
| 6 .. 4 |   | U+F7A3 | wiggleSquareWaveExtraNarrow   |
| 3      |   | U+F7A2 | wiggleSquareWaveVeryNarrow    |
| 2      |   | U+F6B5 | wiggleSquareWaveNarrower      |
| 1      |   | U+EAB7 | wiggleSquareWaveNarrow        |
| 0      |   | U+EAB8 | wiggleSquareWave              |
| -1     |   | U+EAB9 | wiggleSquareWaveWide          |
| -2     |   | U+F6B6 | wiggleSquareWaveWider         |
| :      |                                                                                    |        |                               |
| -6     |  | U+F72B | wiggleSquareWaveQuadrupleWide |

## 'sawtooth

|        |                                                                                     |        |                             |
|--------|-------------------------------------------------------------------------------------|--------|-----------------------------|
| 6 .. 4 |  | U+F7A5 | wiggleSawtoothExtraNarrow   |
| 3      |  | U+F7A4 | wiggleSawtoothVeryNarrow    |
| 2      |  | U+F6B7 | wiggleSawtoothNarrower      |
| 1      |  | U+EABA | wiggleSawtoothNarrow        |
| 0      |  | U+EABB | wiggleSawtooth              |
| -1     |  | U+EABC | wiggleSawtoothWide          |
| -2     |  | U+F6B8 | wiggleSawtoothWider         |
| :      |                                                                                     |        |                             |
| -6     |  | U+F72F | wiggleSawtoothQuadrupleWide |

## Examples

```

\relative c'' {
  \ekmSmuflOn #'textspan

  c1 \ekmStartSpan
    #'vibrato
    #'(3 . -3)
    ##f
  c c c2 \stopTextSpan

  d2 \ekmStartSpan
    #'vibrato-large
    #'(3 . -3)
    #0
  d1 d d \stopTextSpan

  c1 \ekmStartSpan
    #'circular
    #'(-4 . -1)
    ##f
  c d \break d e e f \stopTextSpan

  \textSpannerDown
  d1 \ekmStartSpan
    #'wavy
    #'(0 . -6) % this range uses Ekmelos glyphs
    #'((#xE651 #xE652) . #xE653)
  d d d d d \break d d d d d d
  R1 \stopTextSpan
  \textSpannerNeutral

  c1 \ekmStartSpan
    #'beam
    #'(-5 . 5)
    ##f
  c c c \stopTextSpan
}

```

The image displays three staves of musical notation in treble clef, illustrating the application of Ekmelos glyphs and text spans. The first staff (measures 1-10) shows a sequence of notes with various vibrato and circular motion glyphs above them. The second staff (measures 11-20) features a wavy line below the notes, indicating a specific text span. The third staff (measures 21-30) shows a beam over a group of notes, with a wavy line below the staff. The notation includes various Ekmelos glyphs such as vibrato, circular motion, and wavy lines, which are used to represent specific musical effects or text spans.

# Trill Spans and Pitches

```
\ekmSmuflOn #'trill
```

Draw trill spans assembled from SMuFL multi-segment glyphs, and SMuFL trill pitches.

```
\ekmStartTrillSpan TEMPO
```

Start a trill span with the style 'trill. See also [Multi-segment spanner](#). [LP]

TEMPO is a number or a pair of numbers (rounded to integer) for the segments of the spanner, usually in the range 4 to -4.

```
\startTrillSpan is equivalent to \ekmStartTrillSpan #0
and to \ekmStartSpan #'trill #0 ##f
```

```
\ekmPitchedTrill NOTEHEAD-STYLE PARENS-STYLE MAIN-NOTE AUXILIARY-NOTE
```

Draw a trill pitch. [Err]

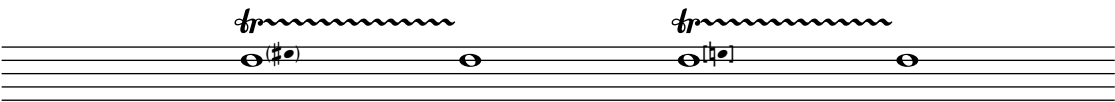
NOTEHEAD-STYLE: See [Note Heads](#).

For variable accidentals on AUXILIARY-NOTE see [Cosmuflily](#).

```
\pitchedTrill is equivalent to \ekmPitchedTrill #'default #'default
```

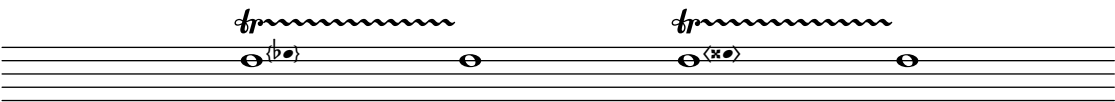
## PARENS-STYLE

|          |        |                        |
|----------|--------|------------------------|
| 'default | U+E26A | accidentalParensLeft   |
|          | U+E26B | accidentalParensRight  |
| 'bracket | U+E26C | accidentalBracketLeft  |
|          | U+E26D | accidentalBracketRight |



## with Ekmelos

|        |        |                      |
|--------|--------|----------------------|
| 'brace | U+F6D4 | accidentalBraceLeft  |
|        | U+F6D5 | accidentalBraceRight |
| 'angle | U+F6D6 | accidentalAngleLeft  |
|        | U+F6D7 | accidentalAngleRight |



## Examples

```

\relative c'' {
  \ekmSmuflOn #'trill

  c1 \ekmStartTrillSpan #0
  c c2. d4 \stopTrillSpan

  c1 \ekmStartTrillSpan #-1
  d1 \ekmStartTrillSpan #-2
  e1 \ekmStartTrillSpan #-3
  f1 \ekmStartTrillSpan #-4
  g1 \stopTrillSpan

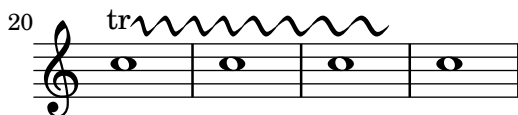
  c,1 \ekmStartTrillSpan #'(-1 . -4)
  d \break e f g \stopTrillSpan

  \afterGrace
  d2 \ekmStartTrillSpan #1
  { c16[ d] }
  c2 \stopTrillSpan

  \ekmPitchedTrill #'triangle #'bracket
  d1 \ekmStartTrillSpan #4
  e
  c \stopTrillSpan

  c1 \ekmStartSpan
    #'trill-vibrato-large
    #'(3 . -3)
    #"tr"
  c c \break c c c c \stopTrillSpan
}

```



Draws a trill with a SMuFL accidental in three ways.

```

ekmUse = #'(trill)
\include "cosmufl.ily"

\relative c'' {
  % use markup
  c1 \ekmStartSpan
    #'trill
    #'(-3 . 2)
    #'(, (markup #:concat
      (:ekm-char #xE566
        #:general-align Y -0.7
        #:fontsize -2
        #:ekm-char #xE262))
      . 0)
  c2. d4 \stopTrillSpan

  % use a ligature
  c1 \ekmStartSpan
    #'trill
    #'(-3 . 2)
    #'((#xE262 #xE566) . 0)
  c2. d4 \stopTrillSpan

  % use a Cosmuflily command
  c1 \ekmStartTrillSpanAccidental
    #'(-3 . 2)
    #1/2
  c2. d4 \stopTrillSpan
}

```



# Ottavation

The ottavation style can be set with

```
\set Staff.ottavationMarkups = #(ekm-ottavation STYLE)
```

## STYLE

|          |    |        |              |
|----------|----|--------|--------------|
| 'numbers | ±1 | U+E510 | ottava       |
|          | ±2 | U+E514 | quindicesima |
|          | ±3 | U+E517 | ventiduesima |

A musical staff with a treble clef and a key signature of one flat. The staff contains a sequence of eighth notes. Above the staff, the numbers 8, 15, and 22 are placed above groups of notes, with dashed lines extending from them. Below the staff, the same numbers 8, 15, and 22 are placed below groups of notes, also with dashed lines.

|           |    |        |                   |
|-----------|----|--------|-------------------|
| 'ordinals | 1  | U+E511 | ottavaAlta        |
|           | -1 | U+E512 | ottavaBassa       |
|           | 2  | U+E515 | quindicesimaAlta  |
|           | -2 | U+E516 | quindicesimaBassa |
|           | 3  | U+E518 | ventiduesimaAlta  |
|           | -3 | U+E519 | ventiduesimaBassa |

A musical staff with a treble clef and a key signature of one flat. The staff contains a sequence of eighth notes. Above the staff, the marks 8va, 15ma, and 22ma are placed above groups of notes, with dashed lines extending from them. Below the staff, the same marks 8va, 15ma, and 22ma are placed below groups of notes, also with dashed lines.

|                  |    |        |                     |
|------------------|----|--------|---------------------|
| 'simple-ordinals | 1  | U+E510 | ottava              |
|                  |    | U+EC97 | octaveBaselineV     |
|                  |    | U+EC91 | octaveBaselineA     |
|                  | -1 | U+E51C | ottavaBassaVb       |
|                  |    | U+E514 | quindicesima        |
|                  |    | U+EC95 | octaveBaselineM     |
|                  | 2  | U+EC91 | octaveBaselineA     |
|                  |    | U+E51D | quindicesimaBassaMb |
|                  |    | U+E517 | ventiduesima        |
|                  | 3  | U+EC95 | octaveBaselineM     |
|                  |    | U+EC91 | octaveBaselineA     |
|                  |    | U+E51E | ventiduesimaBassaMb |

A musical staff with a treble clef and a key signature of one flat. The staff contains a sequence of eighth notes. Above the staff, the marks 8va, 15ma, and 22ma are placed above groups of notes, with dashed lines extending from them. Below the staff, the marks 8vb, 15mb, and 22mb are placed below groups of notes, also with dashed lines.

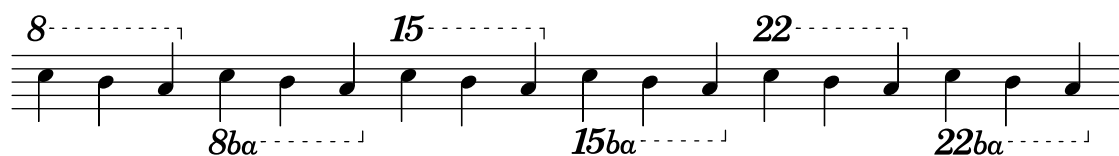
|             |    |        |                     |
|-------------|----|--------|---------------------|
| 'ordinals-b | 1  | U+E511 | ottavaAlta          |
|             | -1 | U+E51C | ottavaBassaVb       |
|             | 2  | U+E515 | quindicesimaAlta    |
|             | -2 | U+E51D | quindicesimaBassaMb |
|             | 3  | U+E518 | ventiduesimaAlta    |
|             | -3 | U+E51E | ventiduesimaBassaMb |

A musical staff with a treble clef and a key signature of one flat. The staff contains a sequence of eighth notes. Above the staff, the marks 8va, 15ma, and 22ma are placed above groups of notes, with dashed lines extending from them. Below the staff, the marks 8vb, 15mb, and 22mb are placed below groups of notes, also with dashed lines.

|                 |    |        |                   |
|-----------------|----|--------|-------------------|
| 'ordinals-bassa | 1  | U+E511 | ottavaAlta        |
|                 | -1 | U+E512 | ottavaBassa       |
|                 |    | U+E51F | octaveBassa       |
|                 | 2  | U+E515 | quindicesimaAlta  |
|                 | -2 | U+E516 | quindicesimaBassa |
|                 |    | U+E51F | octaveBassa       |
|                 | 3  | U+E518 | ventiduesimaAlta  |
|                 | -3 | U+E519 | ventiduesimaBassa |
|                 |    | U+E51F | octaveBassa       |



|             |    |        |                 |
|-------------|----|--------|-----------------|
| 'numbers-ba | 1  | U+E510 | ottava          |
|             | -1 | U+E513 | ottavaBassaBa   |
|             | 2  | U+E514 | quindicesima    |
|             | -2 | U+E514 | quindicesima    |
|             |    | U+EC93 | octaveBaselineB |
|             |    | U+EC91 | octaveBaselineA |
|             | 3  | U+E517 | ventiduesima    |
|             | -3 | U+E517 | ventiduesima    |
|             |    | U+EC93 | octaveBaselineB |
|             |    | U+EC91 | octaveBaselineA |



|              |    |        |                  |
|--------------|----|--------|------------------|
| 'ordinals-ba | 1  | U+E511 | ottavaAlta       |
|              | -1 | U+E513 | ottavaBassaBa    |
|              | 2  | U+E515 | quindicesimaAlta |
|              | -2 | U+E514 | quindicesima     |
|              |    | U+EC93 | octaveBaselineB  |
|              |    | U+EC91 | octaveBaselineA  |
|              | 3  | U+E518 | ventiduesimaAlta |
|              | -3 | U+E517 | ventiduesima     |
|              |    | U+EC93 | octaveBaselineB  |
|              |    | U+EC91 | octaveBaselineA  |



Note: According to the implementation notes of SMuFL Octaves, the suffixes *vb* and *mb* as in 'simple-ordinals and 'ordinals-b are corruptions of the more correct forms *va bassa* and *ma bassa* as in 'ordinals-bassa. The recommended abbreviation for *8va bassa* is *8ba* as in 'numbers-ba and 'ordinals-ba.

with Ekmelos

' numbers

```
'ordinals
```

...

29<sup>ma</sup>

29<sup>ma</sup>

```
'simple-ordinals
```

...

*29<sub>ma</sub>*

*29<sub>mb</sub>*

```
'ordinals-b
```

...

*29<sup>ma</sup>*.....

*29<sup>mb</sup>*.....

```
'ordinals-bassa
```

...

*29ma*

*29ma bassa*

```
'numbers-ba
```

...

29

29<sub>ba</sub>

'ordinals-ba

...

*29<sup>ma</sup>*

*29<sup>ba</sup>*

\ekm-ottavation DEFINITION

Draw an ottavation text according to [DEFINITION](#) as markup.

|                  |                        |        |                     |
|------------------|------------------------|--------|---------------------|
| 8                | <b>8</b>               | U+E510 | ottava              |
| 8 <sup>va</sup>  | <b>8<sup>va</sup></b>  | U+E511 | ottavaAlta          |
| 8 <sub>va</sub>  | <b>8<sub>va</sub></b>  | U+E512 | ottavaBassa         |
| 8 <sub>ba</sub>  | <b>8<sub>ba</sub></b>  | U+E513 | ottavaBassaBa       |
| 8 <sub>vb</sub>  | <b>8<sub>vb</sub></b>  | U+E51C | ottavaBassaVb       |
| 15               | <b>15</b>              | U+E514 | quindicesima        |
| 15 <sup>ma</sup> | <b>15<sup>ma</sup></b> | U+E515 | quindicesimaAlta    |
| 15 <sub>ma</sub> | <b>15<sub>ma</sub></b> | U+E516 | quindicesimaBassa   |
| 15 <sub>mb</sub> | <b>15<sub>mb</sub></b> | U+E51D | quindicesimaBassaMb |
| 22               | <b>22</b>              | U+E517 | ventiduesima        |
| 22 <sup>ma</sup> | <b>22<sup>ma</sup></b> | U+E518 | ventiduesimaAlta    |
| 22 <sub>ma</sub> | <b>22<sub>ma</sub></b> | U+E519 | ventiduesimaBassa   |
| 22 <sub>mb</sub> | <b>22<sub>mb</sub></b> | U+E51E | ventiduesimaBassaMb |
| (                | <b>(</b>               | U+E51A | octaveParensLeft    |
| )                | <b>)</b>               | U+E51B | octaveParensRight   |
| bassa            | <b>bassa</b>           | U+E51F | octaveBassa         |
| loco             | <b>loco</b>            | U+EC90 | octaveLoco          |
| <sup>a</sup>     | <b><sup>a</sup></b>    | U+EC92 | octaveSuperscriptA  |
| <sup>b</sup>     | <b><sup>b</sup></b>    | U+EC94 | octaveSuperscriptB  |
| <sup>m</sup>     | <b><sup>m</sup></b>    | U+EC96 | octaveSuperscriptM  |
| <sup>v</sup>     | <b><sup>v</sup></b>    | U+EC98 | octaveSuperscriptV  |
| a                | <b>a</b>               | U+EC91 | octaveBaselineA     |
| b                | <b>b</b>               | U+EC93 | octaveBaselineB     |
| m                | <b>m</b>               | U+EC95 | octaveBaselineM     |
| v                | <b>v</b>               | U+EC97 | octaveBaselineV     |

with Ekmelos

|                  |                        |        |                         |
|------------------|------------------------|--------|-------------------------|
| 8 <sup>vb</sup>  | <b>8<sup>vb</sup></b>  | U+F652 | ottavaBassaSupVb        |
| 15 <sup>mb</sup> | <b>15<sup>mb</sup></b> | U+F653 | quindicesimaBassaSupMb  |
| 22 <sup>mb</sup> | <b>22<sup>mb</sup></b> | U+F654 | ventiduesimaBassaSupMb  |
| 29               | <b>29</b>              | U+F6F8 | ventinovesima           |
| 29 <sup>ma</sup> | <b>29<sup>ma</sup></b> | U+F6F9 | ventinovesimaAlta       |
| 29 <sub>ma</sub> | <b>29<sub>ma</sub></b> | U+F6FA | ventinovesimaBassa      |
| 29 <sup>mb</sup> | <b>29<sup>mb</sup></b> | U+F655 | ventinovesimaBassaSupMb |
| 29 <sub>mb</sub> | <b>29<sub>mb</sub></b> | U+F6FB | ventinovesimaBassaMb    |

## Tuplet Numbers

---

`\ekmSmuflOn #'tuplet`

Draw SMuFL tuplet numbers as numerator only. Set the first formatting function below, so this switch is not required if one of these functions is set explicitly.

`:` `:` `U+E88A` `tupletColon`

See `\ekm-number` for tuplet digits.

### Tuplet formatting functions

`ekm-tuplet-number::calc-denominator-text`

`ekm-tuplet-number::calc-fraction-text`

`(ekm-tuplet-number::non-default-tuplet-denominator-text NUM)`

`(ekm-tuplet-number::non-default-tuplet-fraction-text NUM DENOM)`

`(ekm-tuplet-number::append-note-wrapper  
FUNCTION DURATION)`

`(ekm-tuplet-number::fraction-with-notes  
NUM-DURATION DENOM-DURATION)`

`(ekm-tuplet-number::non-default-fraction-with-notes  
NUM NUM-DURATION DENOM DENOM-DURATION)`

The last three functions draw 'metronome style notes for the specified durations.

`(ekm-tuplet-number NUM DENOM)`

Draw NUM:DENOM, or NUM only if DENOM is 0. Use the actual tuplet fraction for NUM or DENOM if `#f` is specified. It is called by the first four functions above, i.e. they are equivalent to:

```
(ekm-tuplet-number #f 0)
(ekm-tuplet-number #f #f)
(ekm-tuplet-number NUM 0)
(ekm-tuplet-number NUM DENOM)
```

## Examples

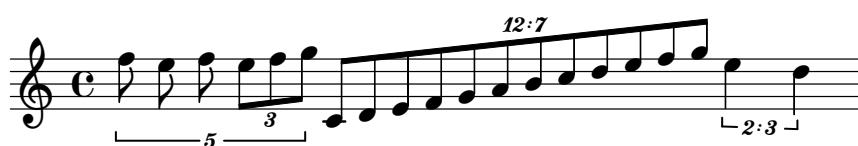
```

\relative c'' {
  \cadenzaOn

  \override TupletNumber.text = #ekm-tuplet-number::calc-denominator-text
  \tuplet 5/4 {
    f8 e f
    \tuplet 3/2 { e[ f g] }
  }

  \override TupletNumber.text = #ekm-tuplet-number::calc-fraction-text
  \tuplet 12/7 { c,,8[ d e f g a b c d e f g] }
  \tuplet 2/3 { e4 d }
}

```



```

\relative c'' {
  \cadenzaOn

  \once \override TupletNumber.text =
    # (ekm-tuplet-number::append-note-wrapper
      ekm-tuplet-number::calc-fraction-text
      (ly:make-duration 2 0))
  \tuplet 5/4 { c8[ d c d c d c d c d] }

  \once \override TupletNumber.text =
    # (ekm-tuplet-number::fraction-with-notes
      (ly:make-duration 2 1)
      (ly:make-duration 3 0))
  \tuplet 3/2 { c4. b a g }

  \once \override TupletNumber.text =
    # (ekm-tuplet-number::non-default-fraction-with-notes
      12 (ly:make-duration 3 0)
      4 (ly:make-duration 2 0))
  \tuplet 3/2 { c4. b a g }
}

```



## Fingering Instructions

`\ekmSmuflOn #'fingering`

Draw SMuFL fingering instructions specified with a digit or with `\finger`, as well as right-hand fingerings specified with `\rightHandFinger`, both using `\ekm-finger`.

Note: The `\thumb` command always produces normal LilyPond output. Use `\finger "th"` to draw the corresponding SMuFL glyph.

`\ekm-finger DEFINITION`

Draw a fingering instruction according to [DEFINITION](#) as markup.

If `DEFINITION` has the prefix `*` the italic style of some symbols (digits and brackets) is drawn.

|    |          |        |                                  |
|----|----------|--------|----------------------------------|
| (  | (        | U+ED28 | fingeringLeftParenthesis         |
| )  | )        | U+ED29 | fingeringRightParenthesis        |
| [  | [        | U+ED2A | fingeringLeftBracket             |
| ]  | ]        | U+ED2B | fingeringRightBracket            |
| .  | •        | U+ED2C | fingeringSeparatorMiddleDot      |
| ,  | ◦        | U+ED2D | fingeringSeparatorMiddleDotWhite |
| /  | /        | U+ED2E | fingeringSeparatorSlash          |
| ^^ | ^        | U+ED20 | fingeringSubstitutionAbove       |
| ~  | ~        | U+ED21 | fingeringSubstitutionBelow       |
| -  | -        | U+ED22 | fingeringSubstitutionDash        |
| M  | ⌈        | U+ED23 | fingeringMultipleNotes           |
| th | ☞        | U+E624 | stringsThumbPosition             |
| ht | ⊙        | U+E625 | stringsThumbPositionTurned       |
| T  | T        | U+ED16 | fingeringTUpper                  |
| t  | <i>t</i> | U+ED18 | fingeringTLower                  |
| p  | <i>p</i> | U+ED17 | fingeringPLower                  |
| i  | <i>i</i> | U+ED19 | fingeringILower                  |
| m  | <i>m</i> | U+ED1A | fingeringMLower                  |
| a  | <i>a</i> | U+ED1B | fingeringALower                  |
| c  | <i>c</i> | U+ED1C | fingeringCLower                  |
| x  | <i>x</i> | U+ED1D | fingeringXLower                  |
| e  | <i>e</i> | U+ED1E | fingeringELower                  |
| o  | <i>o</i> | U+ED1F | fingeringOLower                  |
| q  | <i>q</i> | U+ED8E | fingeringQLower                  |
| s  | <i>s</i> | U+ED8F | fingeringSLower                  |
| R  | ⌊        | U+E66E | keyboardPlayWithRH               |
| RE | ⌋        | U+E66F | keyboardPlayWithRHEnd            |
| L  | ⌈        | U+E670 | keyboardPlayWithLH               |
| LE | ⌋        | U+E671 | keyboardPlayWithLHEnd            |

See [\ekm-number](#) for fingering digits.



```

\relative c'' {
  \ekmSmuflOn #'fingering

  \ekmPlayWith #RIGHT ##t c
  \ekmPlayWith #RIGHT ##f g

  \ekmPlayWith #LEFT ##t c
  \ekmPlayWith #LEFT ##f g
}

```



## String Number Indications

```
\ekmSmuflOn #'stringnumber
```

Draw SMuFL string number indications specified with `\NUMBER`, using `\ekm-string-number`.

**Note:** `\romanStringNumbers` overrides the SMuFL switch so that reverting with

`\arabicStringNumbers` produces normal LilyPond output.

```
\ekm-string-number ARG
```

Draw a string number indication as markup.

ARG is a number or string. For a number or a string representing a number, either the precomposed symbol or the number according to `style` with a circle around is drawn. Any other string, e.g. a Roman numeral, is drawn in italic style.

Used property:

- `style ('sans)`, any number style except `'string`.

See `\ekm-number` for string number (guitar string) symbols and for number styles.

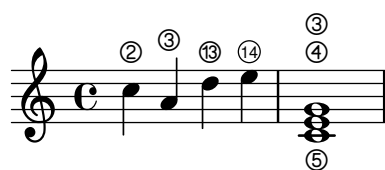
## Examples

```
\relative c' {
  \ekmSmuflOn #'stringnumber
```

```
  c \2
  a \3
  d \13
  e \14
```

```
  < c,\5 e\4 g\3 >1
```

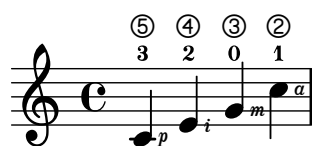
```
}
```



```
\relative c' {
  \ekmSmuflOn #'(fingering stringnumber)
```

```
  < c -3 \5 \rightHandFinger #1 >
  < e -2 \4 \rightHandFinger #2 >
  < g -0 \3 \rightHandFinger #3 >
  < c -1 \2 \rightHandFinger #4 >
```

```
}
```



## Piano Pedals

\ekmSmuflOn #'pedal

Draw SMuFL piano pedals for sustain, sostenuto, and una corda, using \ekm-piano-pedal .

\ekm-piano-pedal DEFINITION

Draw piano pedal symbols according to [DEFINITION](#) as markup.

|       |                                                                                     |        |                              |
|-------|-------------------------------------------------------------------------------------|--------|------------------------------|
| Ped.  |    | U+E650 | keyboardPedalPed             |
| P     |    | U+E651 | keyboardPedalP               |
| e     |    | U+E652 | keyboardPedalE               |
| d     |    | U+E653 | keyboardPedalD               |
| Sost. |    | U+E659 | keyboardPedalSost            |
| S     |    | U+E65A | keyboardPedalS               |
| .     |    | U+E654 | keyboardPedalDot             |
| –     |    | U+E658 | keyboardPedalHyphen          |
| *     |    | U+E655 | keyboardPedalUp              |
| o     |    | U+E65D | keyboardPedalUpSpecial       |
| ,     |   | U+E65B | keyboardPedalHalf2           |
| '     |  | U+E65C | keyboardPedalHalf3           |
| H     |  | U+E656 | keyboardPedalHalf            |
| ^     |  | U+E657 | keyboardPedalUpNotch         |
| l     |  | U+E65E | keyboardLeftPedalPictogram   |
| m     |  | U+E65F | keyboardMiddlePedalPictogram |
| r     |  | U+E660 | keyboardRightPedalPictogram  |
| (     |  | U+E676 | keyboardPedalParensLeft      |
| )     |  | U+E677 | keyboardPedalParensRight     |

with Ekmelos

|           |                                                                                     |        |                               |
|-----------|-------------------------------------------------------------------------------------|--------|-------------------------------|
| Ped       |  | U+F434 | keyboardPedalPedNoDot         |
| ConPed.   |  | U+F79E | keyboardPedalConPed           |
| SenzaPed. |  | U+F79F | keyboardPedalSenzaPed         |
| Sost      |  | U+F435 | keyboardPedalSostNoDot        |
| Sos.      |  | U+F6D1 | keyboardPedalSos2             |
| sos.      |  | U+F6D0 | keyboardPedalSos              |
| unacorda  | <i>una corda</i>                                                                    | U+F6CC | keyboardPedalUnaCorda         |
| tre corde | <i>tre corde</i>                                                                    | U+F6CD | keyboardPedalTreCorde         |
| u.c.      | <i>u.c.</i>                                                                         | U+F6CE | keyboardPedalUC               |
| t.c.      | <i>t.c.</i>                                                                         | U+F6CF | keyboardPedalTC               |
| 1/2Ped    |  | U+F6B0 | keyboardPedalHalf4            |
| 1/4       |  | U+F6BA | keyboardPedalPosQuarter       |
| 1/2       |  | U+F6BB | keyboardPedalPosHalf          |
| 3/4       |  | U+F6BC | keyboardPedalPosThreeQuarters |
| 1         |  | U+F6BD | keyboardPedalPosFull          |

## Examples

```

\new Staff \with {
  \ekmSmuflOn #'pedal
}
\relative c'' {
  \set Staff.pedalSustainStrings = #("Ped." "H" "*")

  c4 \sustainOn d c b
  c \sustainOff \sustainOn d c b
  c1 \sustainOff
}

```



```

\new Staff \with {
  \ekmSmuflOn #'pedal
}
\relative c'' {
  \set Staff.pedalSostenutoStyle = #'text
  \set Staff.pedalSostenutoStrings = #("S-P" "(" "S. *")

  c4 \sostenutoOn d c b
  c \sostenutoOff \sostenutoOn d c b
  c1 \sostenutoOff
}

```



```

\new Staff \with {
  \ekmSmuflOn #'pedal
}
\relative c'' {
  % draws Ekmelos glyphs "unacorda" and "t.c."
  \set Staff.pedalUnaCordaStyle = #'text
  \set Staff.pedalUnaCordaStrings = #("unacorda" "^__t.c." "o")

  c4 \unaCorda d c b
  c \treCorde \unaCorda d c b
  c1 \treCorde
}

```



## Harp Pedals

`\ekm-harp-pedal` DEFINITION

Draw a harp pedal diagram according to [DEFINITION](#) as markup. Space between tokens is ignored.

|                 |                                                                                   |        |                   |
|-----------------|-----------------------------------------------------------------------------------|--------|-------------------|
| <code>^</code>  |  | U+E680 | harpPedalRaised   |
| <code>o^</code> |  | U+E681 | harpPedalCentered |
| <code>-</code>  |                                                                                   |        |                   |
| <code>o-</code> |                                                                                   |        |                   |
| <code>v</code>  |  | U+E682 | harpPedalLowered  |
| <code>ov</code> |                                                                                   |        |                   |
| <code> </code>  |  | U+E683 | harpPedalDivider  |

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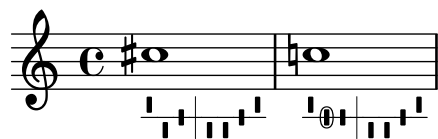
|                 |                                                                                   |        |                         |
|-----------------|-----------------------------------------------------------------------------------|--------|-------------------------|
| <code>o^</code> |  | U+F648 | harpPedalRaisedChange   |
| <code>o-</code> |  | U+F649 | harpPedalCenteredChange |
| <code>ov</code> |  | U+F64A | harpPedalLoweredChange  |

`\ekm-harp-change` EXTEXT Y

Draw [EXTEXT](#) as markup with an ellipse translated vertically by Y. Used by change tokens such as `o^`.

## Examples

```
\relative c'' {
  \textLengthOn
  cis1 _ \markup \ekm-harp-pedal #"^ v - | v v - ^"
  c! _ \markup \ekm-harp-pedal #"^ o- - | v v - ^"
}
```



# Fret Diagrams

---

`\ekm-fret-diagram-terse` DEFINITION

Draw a fret diagram according to [DEFINITION](#) as markup. Fingering is always placed below.

Used properties:

- `fret-diagram-details top-fret-thickness (3)`: A value > 1 draws the ...Nut glyph.
- `fret-diagram-details finger-code (#t) : 'none` draws no fingering.
- `fret-diagram-details finger-style ('sans)`

|           |                                                                                     |        |                       |
|-----------|-------------------------------------------------------------------------------------|--------|-----------------------|
| 3 strings |    | U+E850 | fretboard3String      |
|           |    | U+E851 | fretboard3StringNut   |
| 4 strings |    | U+E852 | fretboard4String      |
|           |    | U+E853 | fretboard4StringNut   |
| 5 strings |    | U+E854 | fretboard5String      |
|           |    | U+E855 | fretboard5StringNut   |
| 6 strings |   | U+E856 | fretboard6String      |
|           |  | U+E857 | fretboard6StringNut   |
| .         |  | U+E858 | fretboardFilledCircle |
| x         |  | U+E859 | fretboardX            |
| o         |  | U+E85A | fretboardO            |

## Examples

```

\relative c'' {
  \textLengthOn

  c1 ^ \markup \ekm-fret-diagram-terse #"x;3-3;2-2;o;1-1;o;"

  cis1 ^ \markup \ekm-fret-diagram-terse #"x;x;3-3;1-1-(;2-2;1-1-);"

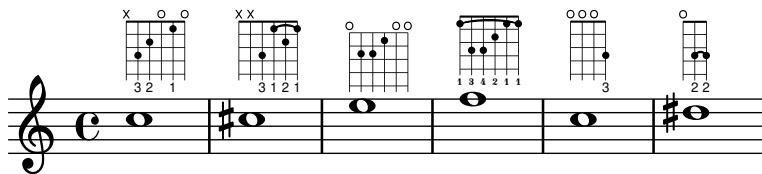
  e1 ^ \markup {
    \override #'(fret-diagram-details . (
      (top-fret-thickness . 1)
      (finger-code . none)))
    \ekm-fret-diagram-terse #"o;2-2;2-3;1-1;o;o;"
  }

  f1 ^ \markup {
    \override #'(fret-diagram-details . (
      (finger-style . fingering)))
    \ekm-fret-diagram-terse #"1-1-(;3-3;3-4;2-2;1-1;1-1-);"
  }

  c1 ^ \markup \ekm-fret-diagram-terse #"o;o;o;3-3;"

  dis1 ^ \markup \ekm-fret-diagram-terse #"o;3-2-(;3-2-);"
}

```



# Accordion Registers

`\ekm-accordion NAME`

Draw an accordion register symbol as markup. Most of the symbols use precomposed glyphs. The others are composed using `accdnCombRH3RanksEmpty` (U+E8C6) etc.

NAME has a prefix for the register style separated by a space. "d" (discant) is the default and can be omitted.

Note: The module `(scm accreg)` is not required.

`\ekmAccordion NAME`

Set an accordion register symbol as a standalone music event. This is equivalent to

```
<> ^ \markup \ekm-accordion NAME
```

## NAME

|          |                                                                                     |        |                                    |
|----------|-------------------------------------------------------------------------------------|--------|------------------------------------|
| "d ..."  | Discant                                                                             |        |                                    |
| "d 1"    |    | U+E8A4 | accdnRH3RanksBassoon               |
| "d 10"   |    | U+E8A1 | accdnRH3RanksClarinet              |
| "d 11"   |    | U+E8AB | accdnRH3RanksBandoneon             |
| "d 1+0"  |  | U+E8A2 | accdnRH3RanksUpperTremolo8         |
| "d 1+1"  |  |        |                                    |
| "d 1-0"  |  | U+E8A3 | accdnRH3RanksLowerTremolo8         |
| "d 1-1"  |  |        |                                    |
| "d 20"   |  | U+E8AE | accdnRH3RanksTwoChoirs             |
| "d 21"   |  | U+E8AF | accdnRH3RanksTremoloLower8ve       |
| "d 2+0"  |  | U+E8A6 | accdnRH3RanksViolin                |
| "d 2+1"  |  | U+E8AC | accdnRH3RanksAccordion             |
| "d 2-0"  |  |        |                                    |
| "d 2-1"  |  |        |                                    |
| "d 30"   |  | U+E8A8 | accdnRH3RanksAuthenticMusette      |
| "d 31"   |  | U+E8B1 | accdnRH3RanksDoubleTremoloLower8ve |
| "d 100"  |  | U+E8A0 | accdnRH3RanksPiccolo               |
| "d 101"  |  | U+E8A9 | accdnRH3RanksOrgan                 |
| "d 110"  |  | U+E8A5 | accdnRH3RanksOboe                  |
| "d 111"  |  | U+E8AA | accdnRH3RanksHarmonium             |
| "d 11+0" |  |        |                                    |
| "d 11+1" |  |        |                                    |
| "d 11-0" |  |        |                                    |
| "d 11-1" |  |        |                                    |

|          |  |        |                                    |
|----------|--|--------|------------------------------------|
| "d 120"  |  | U+E8B0 | accdnRH3RanksTremoloUpper8ve       |
| "d 121"  |  | U+E8AD | accdnRH3RanksMaster                |
| "d 12+0" |  | U+E8A7 | accdnRH3RanksImitationMusette      |
| "d 12+1" |  |        |                                    |
| "d 12-0" |  |        |                                    |
| "d 12-1" |  |        |                                    |
| "d 130"  |  | U+E8B2 | accdnRH3RanksDoubleTremoloUpper8ve |
| "d 131"  |  | U+E8B3 | accdnRH3RanksFullFactory           |

"sb ..." Standard bass

|                 |  |        |                        |
|-----------------|--|--------|------------------------|
| "sb Soprano"    |  | U+E8B4 | accdnRH4RanksSoprano   |
| "sb Alto"       |  | U+E8B5 | accdnRH4RanksAlto      |
| "sb Tenor"      |  | U+E8B6 | accdnRH4RanksTenor     |
| "sb Master"     |  | U+E8B7 | accdnRH4RanksMaster    |
| "sb Soft Bass"  |  | U+E8B8 | accdnRH4RanksSoftBass  |
| "sb Soft Tenor" |  | U+E8B9 | accdnRH4RanksSoftTenor |
| "sb Bass/Alto"  |  | U+E8BA | accdnRH4RanksBassAlto  |

"sb4 ..." Standard bass, four reed

|                      |  |        |                        |
|----------------------|--|--------|------------------------|
| "sb4 Soprano"        |  | U+E8B4 | accdnRH4RanksSoprano   |
| "sb4 Alto"           |  | U+E8B5 | accdnRH4RanksAlto      |
| "sb4 Tenor"          |  |        |                        |
| "sb4 Master"         |  |        |                        |
| "sb4 Soft Bass"      |  |        |                        |
| "sb4 Bass/Alto"      |  | U+E8BA | accdnRH4RanksBassAlto  |
| "sb4 Soft Bass/Alto" |  |        |                        |
| "sb4 Soft Tenor"     |  | U+E8B9 | accdnRH4RanksSoftTenor |

|                      |                                                                                     |        |                                |
|----------------------|-------------------------------------------------------------------------------------|--------|--------------------------------|
| "sb5 ..."            | Standard bass, five reed                                                            |        |                                |
| "sb5 Bass/Alto"      |    | U+E8BA | accdnRH4RanksBassAlto          |
| "sb5 Soft Bass/Alto" |    |        |                                |
| "sb5 Alto"           |    |        |                                |
| "sb5 Tenor"          |    |        |                                |
| "sb5 Master"         |    |        |                                |
| "sb5 Soft Bass"      |    |        |                                |
| "sb5 Soft Tenor"     |    | U+E8B9 | accdnRH4RanksSoftTenor         |
| "sb5 Soprano"        |    | U+E8B4 | accdnRH4RanksSoprano           |
| "sb5 Sopranos"       |    |        |                                |
| "sb5 Solo Bass"      |    |        |                                |
|                      |                                                                                     |        |                                |
| "sb6 ..."            | Standard bass, six reed                                                             |        |                                |
| "sb6 Soprano"        |   | U+E8B4 | accdnRH4RanksSoprano           |
| "sb6 Alto"           |  |        |                                |
| "sb6 Soft Tenor"     |  | U+E8B9 | accdnRH4RanksSoftTenor         |
| "sb6 Master"         |  | U+E8B7 | accdnRH4RanksMaster            |
| "sb6 Alto/Soprano"   |  |        |                                |
| "sb6 Bass/Alto"      |  | U+E8BA | accdnRH4RanksBassAlto          |
| "sb6 Soft Bass"      |  | U+E8B8 | accdnRH4RanksSoftBass          |
|                      |                                                                                     |        |                                |
| "fb ..."             | Free bass                                                                           |        |                                |
| "fb 10"              |  | U+E8BB | accdnLH2Ranks8Round            |
| "fb 1"               |  | U+E8BC | accdnLH2Ranks16Round           |
| "fb 11"              |  | U+E8BD | accdnLH2Ranks8Plus16Round      |
| "fb Master"          |  | U+E8BE | accdnLH2RanksMasterRound       |
| "fb Master 1"        |  | U+E8BF | accdnLH2RanksMasterPlus16Round |
| "fb Master 11"       |  | U+E8C0 | accdnLH2RanksFullMasterRound   |

|          |                                                                                   |        |                            |
|----------|-----------------------------------------------------------------------------------|--------|----------------------------|
| "sq ..." | Square                                                                            |        |                            |
| "sq 1"   |  | U+E8C1 | accdnLH3Ranks8Square       |
| "sq 100" |  | U+E8C2 | accdnLH3Ranks2Square       |
| "sq 2"   |  | U+E8C3 | accdnLH3RanksDouble8Square |
| "sq 101" |  | U+E8C4 | accdnLH3Ranks2Plus8Square  |
| "sq 102" |  | U+E8C5 | accdnLH3RanksTuttiSquare   |

Accordion Ricochet

```
\ekmRicochet NUMBER
```

Draw a ricochet symbol as an expressive mark (script). [\[LP\]](#)

NUMBER

|   |  |        |                |
|---|--|--------|----------------|
| 2 |  | U+E8CD | accdnRicochet2 |
| 3 |  | U+E8CE | accdnRicochet3 |
| 4 |  | U+E8CF | accdnRicochet4 |
| 5 |  | U+E8D0 | accdnRicochet5 |
| 6 |  | U+E8D1 | accdnRicochet6 |

```
\ekmStemRicochet NUMBER MUSIC
```

Draw a ricochet symbol vertically centered on the stems in MUSIC.

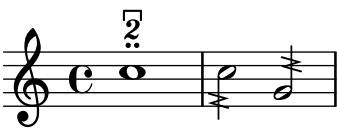
NUMBER

|   |  |        |                    |
|---|--|--------|--------------------|
| 2 |  | U+E8D2 | accdnRicochetStem2 |
| 3 |  | U+E8D3 | accdnRicochetStem3 |
| 4 |  | U+E8D4 | accdnRicochetStem4 |
| 5 |  | U+E8D5 | accdnRicochetStem5 |
| 6 |  | U+E8D6 | accdnRicochetStem6 |

Examples

```
\relative c'' {
  \ekmSmuflOn #'script
  c1 ^ \ekmRicochet #2

  \ekmStemRicochet #3 { c2 g }
}
```



## Figured Bass

---

`\ekmSmuflOn #'fbass`

Draw SMuFL bass figures with `\figuremode`. Some raised/diminished figures use precomposed glyphs which ignore the property `figuredBassPlusDirection`.

|     |                                                                                     |        |                          |
|-----|-------------------------------------------------------------------------------------|--------|--------------------------|
| !   |    | U+EA65 | figbassNatural           |
| –   |    | U+EA64 | figbassFlat              |
| +   |    | U+EA66 | figbassSharp             |
| --  |    | U+EA63 | figbassDoubleFlat        |
| ++  |    | U+EA67 | figbassDoubleSharp       |
| --- |    | U+ECC1 | figbassTripleFlat        |
| +++ |    | U+ECC2 | figbassTripleSharp       |
| \+  |    | U+EA6C | figbassPlus              |
| /   |    | U+EA6D | figbassCombiningRaising  |
| \\  |    | U+EA6E | figbassCombiningLowering |
| 2\+ |    | U+EA53 | figbass2Raised           |
| 4\+ |    | U+EA56 | figbass4Raised           |
| 5\+ |    | U+EA58 | figbass5Raised1          |
| 5\\ |  | U+EA59 | figbass5Raised2          |
| 5/  |  | U+EA5A | figbass5Raised3          |
| 6\\ |  | U+EA5C | figbass6Raised           |
| 6\+ |  | U+EA6F | figbass6Raised2          |
| 7\+ |  | U+EA5E | figbass7Raised1          |
| 7\\ |  | U+EA5F | figbass7Raised2          |
| 7/  |  | U+ECC0 | figbass7Diminished       |
| 9\\ |  | U+EA62 | figbass9Raised           |

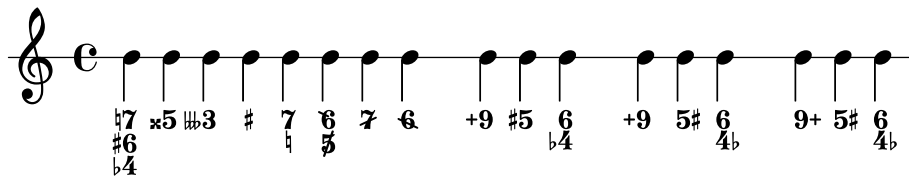
See [\ekm-number](#) for figured bass digits.

## Examples

```

\new Staff
<<
  \relative c'' {
    \cadenzaOn
    b4 b b b b b b s
    b b b s
    b b b s
    b b b
  }
  \figures {
    <7! 6+ 4-> <5++> <3---> <_+> <7 _!> <6\+ 5/> <7/> <6\\> r
    <9\+> <5+> <6 4-> r
    \set figuredBassAlterationDirection = #RIGHT
    <9\+> <5+> <6 4-> r
    \set figuredBassPlusDirection = #RIGHT
    <9\+> <5+> <6 4->
  }
>>
\layout {
  \context {
    \Score
    \ekmSmuflOn #'fbass
    \override StaffSymbol.line-count = #1
  }
}

```



## Lyrics

`\ekmSmuflOn #'lyric`

Draw the words in a lyric input mode (`\lyricmode` etc.) with `\ekm-tied-lyric`.

Note: The characters `_` and `%` must be quoted in order to be passed on to this command.

`\ekm-tied-lyric STRING`

Draw `STRING` as markup, replacing tokens with the corresponding glyphs. The space between the adjoining words depends on the width of the respective glyph, while the property `word-space` is ignored.

In the token `~?~` for narrow elision, `?` can be any Unicode character, not only ASCII.

|     |   |        |                      |
|-----|---|--------|----------------------|
| ~   | ⵿ | U+E551 | lyricsElision        |
| ~?~ | ⵿ | U+E550 | lyricsElisionNarrow  |
| ~~  | ⵿ | U+E552 | lyricsElisionWide    |
| —   | — | U+E553 | lyricsHyphenBaseline |
| %   | ⵿ | U+E555 | lyricsTextRepeat     |

### Example

```
\relative {
  \cadenzaOn
  b'~ b c fis, fis c' b e,
}
\addlyrics {
  Che~~in ques -- ta~ē~in quel -- l'al -- "tr_on" -- "da %"
}
\layout {
  \context {
    \Score
    \ekmSmuflOn #'lyric
  }
}
```



## Chord Name Symbols

---

`\ekmSmuflOn #'chord`

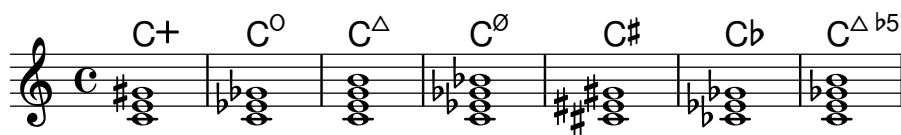
Draw SMuFL chord name symbols.

|       |   |        |                    |
|-------|---|--------|--------------------|
| dim   | ◯ | U+E870 | csymDiminished     |
| m7.5- | ∅ | U+E871 | csymHalfDiminished |
| maj7  | Δ | U+E873 | csymMajorSeventh   |
| aug   | + | U+E872 | csymAugmented      |
| -     | − | U+E874 | csymMinor          |

### Examples

```
someChords = \chordmode {
  c1:aug
  c:dim
  c:maj7
  c:m7.5-
  % use cosmufl.ily to draw SMuFL accidentals
  cis
  ces
  <c' e' ges' b'>
}

<<
  \new ChordNames { \someChords }
  \someChords
>>
\layout {
  \context {
    \Score
    \ekmSmuflOn #'chord
  }
}
```



## Analytics Symbols

---

\ekm-analytics DEFINITION

Draw analytics symbols according to [DEFINITION](#) as markup.

|     |                                                                                     |        |                                   |
|-----|-------------------------------------------------------------------------------------|--------|-----------------------------------|
| H   |    | U+E860 | analyticsHauptstimme              |
| CH  |    | U+E86A | analyticsChoralmelodie            |
| RH  |    | U+E86B | analyticsHauptrhythmus            |
| N   |    | U+E861 | analyticsNebenstimme              |
| [   |    | U+E862 | analyticsStartStimme              |
| ]   |    | U+E863 | analyticsEndStimme                |
| Th  |    | U+E864 | analyticsTheme                    |
| hT  |    | U+E865 | analyticsThemeRetrograde          |
| ihT |    | U+E866 | analyticsThemeRetrogradeInversion |
| iTh |    | U+E867 | analyticsThemeInversion           |
| T   |    | U+E868 | analyticsTheme1                   |
| iT  |  | U+E869 | analyticsInversion1               |

Function Theory Symbols

`\ekm-func` DEFINITION

Draw a function theory symbol according to DEFINITION as markup. The definition string consists of several parts which are all optional:

    Paren Function , Bass , Soprano ^ Extra ... Paren

The bass symbol is placed below the function symbol.

The soprano symbol is placed above the function symbol.

The extra symbols are stacked vertically and raised to the right of the function symbol.

A leading/trailing parenthesis, bracket, or brace is placed separately before/after the entire symbol.

Used properties:

- `font-size` (0) for the function symbol.
- `func-size` (-4) relative to the font size for bass, soprano, and extra symbols.
- `func-skip` (2.5) for vertical distances.
- `func-space` (0.3) for horizontal space around the function symbol.

`\ekmFunc` DEFINITION

Set a function theory symbol as a music expression, for use in a `Lyrics` context. The symbol is drawn with a 4 steps smaller font size compared to `\ekm-func`.

DEFINITION is a string as described above, with a further optional suffix:

- Starts an extender line after the symbol.
- . Stops an extender line at the symbol.
- + Inserts the symbol between notes with `\set stanza`.
- \* Dito but with the 4 steps larger font size of `\ekm-func`.

Note that the `Lyrics` context requires the `Text_spanner_engraver` to draw extender lines.

`\ekmFuncList` DEFINITION-LIST

Set a sequence of function theory symbols as music expressions, for use in a `Lyrics` context.

DEFINITION-LIST is a list of strings as for `\ekmFunc`.

|    |                |        |                |
|----|----------------|--------|----------------|
| T  | T              | U+EA8B | functionTUpper |
| Tg | T <sub>g</sub> |        |                |
| Tp | T <sub>p</sub> |        |                |
| t  | t              | U+EA8C | functionTLower |
| D  | D              | U+EA7F | functionDUpper |
| /D | ∅              |        |                |
| Dp | D <sub>p</sub> |        |                |

|     |    |        |                     |
|-----|----|--------|---------------------|
| DD  | Ḑ  | U+EA81 | functionDD          |
| /DD | Ḑ̄ | U+EA82 | functionSlashedDD   |
| d   | ḑ  | U+EA80 | functionDLower      |
| S   | Ṡ  | U+EA89 | functionSUpper      |
| Sg  | Ṡḡ |        |                     |
| Sp  | Ṡḫ |        |                     |
| SS  | Ṣ  | U+EA7D | functionSSUpper     |
| s   | ṣ  | U+EA8A | functionSLower      |
| ss  | ṣṣ | U+EA7E | functionSSLower     |
| F   | Ḟ  | U+EA99 | functionFUpper      |
| G   | Ḡ  | U+EA83 | functionGUpper      |
| g   | ḡ  | U+EA84 | functionGLower      |
| I   | Ḩ  | U+EA9A | functionIUpper      |
| i   | ḩ  | U+EA9B | functionILower      |
| K   | Ḳ  | U+EA9C | functionKUpper      |
| k   | ḳ  | U+EA9D | functionKLower      |
| L   | Ḵ  | U+EA9E | functionLUpper      |
| l   | ḵ  | U+EA9F | functionLLower      |
| M   | Ḷ  | U+ED00 | functionMUpper      |
| m   | ḷ  | U+ED01 | functionMLower      |
| N   | Ḹ  | U+EA85 | functionNUpper      |
| n   | ḹ  | U+EA86 | functionNLower      |
| P   | Ṁ  | U+EA87 | functionPUpper      |
| p   | ṁ  | U+EA88 | functionPLower      |
| r   | Ṛ  | U+ED03 | functionRLower      |
| V   | Ṫ  | U+EA8D | functionVUpper      |
| v   | ṫ  | U+EA8E | functionVLower      |
| <   | Ṽ  | U+EA7A | functionLessThan    |
| >   | Ṽ̄ | U+EA7C | functionGreaterThan |
| -   | Ṽ̄ | U+EA7B | functionMinus       |
| +   | Ṽ̄ | U+EA98 | functionPlus        |
| o   | Ṽ̄ | U+EA97 | functionRing        |

|     |     |        |                           |
|-----|-----|--------|---------------------------|
| (   | (   | U+EA91 | functionParensLeft        |
| )   | )   | U+EA92 | functionParensRight       |
| [   | [   | U+EA8F | functionBracketLeft       |
| ]   | ]   | U+EA90 | functionBracketRight      |
| {   | <   | U+EA93 | functionAngleLeft         |
| }   | >   | U+EA94 | functionAngleRight        |
| ..  | ..  | U+EA95 | functionRepetition1       |
| ..+ | ..+ | U+EA96 | functionRepetition2       |
| b   | b   | U+ED60 | csymAccidentalFlat        |
| #   | #   | U+ED62 | csymAccidentalSharp       |
| bb  | bb  | U+ED64 | csymAccidentalDoubleFlat  |
| x   | x   | U+ED63 | csymAccidentalDoubleSharp |
| =   | =   | U+ED61 | csymAccidentalNatural     |
| ~   |     |        |                           |

with Ekmelos

|    |   |        |                  |
|----|---|--------|------------------|
| /D | ∅ | U+F644 | functionSlashedD |
|----|---|--------|------------------|

b # bb x = draw the standard accidentals for chord symbols.

~ draws a space with the dimensions of functionZero (U+EA70). This is useful for empty extra symbols.

See [\ekm-number](#) for function theory digits.

## Example

Uses `\ekm-func` in text scripts to attach function theory symbols to chords and spacer rest. It sets `\textLengthOn` and `TextScript.staff-padding` for a consistent vertical alignment.

```
\relative c' {
  \textLengthOn

  \override TextScript.staff-padding = #6
  <c e g bes>2 _ \markup \ekm-func "D^7 "
  <e g bes! c> _ \markup \ekm-func "(D,3^7) "

  \override TextScript.staff-padding = #11
  <c e g c>4 _ \markup \ekm-func "T____"
  <g e' g c> _ \markup \ekm-func "D^4^6"
  s _ \markup \ekm-func "^-^-"
  <g d' g b> _ \markup \ekm-func "^3^5"

  \key es \major
  \override TextScript.staff-padding = #7
  <g' b d>1 _ \markup \ekm-func "V#"
  <f as c e> _ \markup \ekm-func "IV^7#"
  <ces es as!> _ \markup \ekm-func "VI,b"
}
```

The image displays a musical staff in treble clef with a key signature of one flat (B-flat) and a common time signature (C). The staff contains six measures of music, each with a chord and a corresponding function theory symbol below it. The chords and symbols are: 1. D7 (D7) with a subscript 3; 2. T (T); 3. D (D) with a subscript 4-3; 4. V# (V#); 5. IV7# (IV7#); 6. VI (VI) with a subscript b. The function theory symbols are: 1. D7 (D7) with a subscript 3; 2. T (T); 3. D (D) with a subscript 4-3; 4. V# (V#); 5. IV7# (IV7#); 6. VI (VI) with a subscript b.

## Example

Uses `\ekmFuncList` in a `Lyrics` context to synchronise function theory symbols to music. The `Lyrics` context requires `Text_spanner_engraver` and is aligned to a `NullVoice` context.

It is taken from [lsr.di.unimi.it/LSR/Item?id=967](http://lsr.di.unimi.it/LSR/Item?id=967) and adapted for Esmuflily.

```
funcSoprano = \relative c'' {
  e4 e e( d)
  c4 d d2
  d4 e8 d c4 c
  d8( c) <b g>4 c2
}

funcAltTenor = \relative c'' {
  <c g>4 <bes g> <a f>2
  <a d,>4 <c a> <c a>( <b g>)
  <b e,>2 <g e>4 <a f>
  <a d,>4 d,8( f) <g e>2
}

funcBass = \relative c {
  \clef bass
  c4 cis d2
  f4 fis g2
  gis2 bes4 a8 g
  fis4 g c,2
}

funcAligner = \relative c {
  c4 cis d d
  f4 fis g g
  gis4 gis8 gis bes4 a8 g
  fis8 fis g g c,2
}

funcSymbols = \lyricmode {
  \set stanza = #"C major:"
  \ekmFuncList #'(
    "T,,3" " (*" "/D,3^7^9>" ")*" "Sp^9-" "^8."
    "S^5^6" "(D,3^7)" "D^2^4-" "^1^3."
    "(D,3^7-" "^8" "^7." "_" [Tp] +" "(D,7)" "S,3-" " ,2."
    "DD,3^8-" "^7." "D^5-" "^7." "T"
  )
}

\layout {
  \context {
    \Lyrics
    \consists "Text_spanner_engraver"
    \override StanzaNumber.font-family = #'sans
    \override StanzaNumber.font-series = #'medium
  }
}
```

```

\new GrandStaff
<<
  \new Staff
    \new Voice \partCombine \funcSoprano \funcAltTenor

  \new Staff
    <<
      \new Voice \funcBass
      \new NullVoice = "funcaligner" \funcAligner
      \new Lyrics \lyricsto "funcaligner" \funcSymbols
    >>
  >>

```

C major:  $\overset{3}{T} \left( \overset{9}{\underset{3}{D^7}} \right) S_p \overset{9}{\text{—}} \overset{8}{\text{—}} \overset{6}{S^5} \left( \overset{4}{\underset{3}{D^7}} \right) \overset{3}{D^2} \overset{1}{\text{—}} \left( \overset{7}{\underset{3}{D^7}} \overset{8}{\text{—}} \overset{7}{\text{—}} \right) [Tp] \left( \overset{7}{D} \right) \overset{3}{S} \overset{2}{\text{—}} \overset{8}{\underset{3}{D}} \overset{7}{\text{—}} \overset{5}{D} \overset{7}{\text{—}} T$

## Percussion Symbols

---

\ekm-beater STYLE ORIENTATION

Draw a percussion beater of STYLE according to [ORIENTATION](#) as markup.

Percussion beaters can have glyphs for several orientations. The remaining orientations are achieved by flipping or rotating through 90° or 30°. Most of the following styles have glyphs for the orientations N, S, NE, NW or N, S.

### STYLE

|                 |                                                                                     |        |                              |
|-----------------|-------------------------------------------------------------------------------------|--------|------------------------------|
| 'xyl-soft       |    | U+E770 | pictBeaterSoftXylophoneUp    |
| 'xyl-medium     |    | U+E774 | pictBeaterMediumXylophoneUp  |
| 'xyl-hard       |    | U+E778 | pictBeaterHardXylophoneUp    |
| 'xyl-wood       |    | U+E77C | pictBeaterWoodXylophoneUp    |
| 'glsp-soft      |    | U+E780 | pictBeaterSoftGlockenspielUp |
| 'glsp-hard      |    | U+E784 | pictBeaterHardGlockenspielUp |
| 'timpani-soft   |    | U+E788 | pictBeaterSoftTimpaniUp      |
| 'timpani-medium |    | U+E78C | pictBeaterMediumTimpaniUp    |
| 'timpani-hard   |   | U+E790 | pictBeaterHardTimpaniUp      |
| 'timpani-wood   |  | U+E794 | pictBeaterWoodTimpaniUp      |
| 'yarn-soft      |  | U+E7A2 | pictBeaterSoftYarnUp         |
| 'yarn-medium    |  | U+E7A6 | pictBeaterMediumYarnUp       |
| 'yarn-hard      |  | U+E7AA | pictBeaterHardYarnUp         |
| 'gum-soft       |  | U+E7BB | pictGumSoftUp                |
| 'gum-medium     |  | U+E7BF | pictGumMediumUp              |
| 'gum-hard       |  | U+E7C3 | pictGumHardUp                |
| 'bass-soft      |  | U+E798 | pictBeaterSoftBassDrumUp     |
| 'bass-medium    |  | U+E79A | pictBeaterMediumBassDrumUp   |
| 'bass-hard      |  | U+E79C | pictBeaterHardBassDrumUp     |
| 'bass-metal     |  | U+E79E | pictBeaterMetalBassDrumUp    |
| 'bass-double    |  | U+E7A0 | pictBeaterDoubleBassDrumUp   |
| 'stick          |  | U+E7E8 | pictDrumStick                |
| 'stick-snare    |  | U+E7D1 | pictBeaterSnareSticksUp      |
| 'stick-jazz     |  | U+E7D3 | pictBeaterJazzSticksUp       |
| 'hammer-wood    |  | U+E7CB | pictBeaterHammerWoodUp       |
| 'hammer-plastic |  | U+E7CD | pictBeaterHammerPlasticUp    |
| 'hammer-metal   |  | U+E7CF | pictBeaterHammerMetalUp      |

|                 |                                                                                   |        |                          |
|-----------------|-----------------------------------------------------------------------------------|--------|--------------------------|
| 'wound-hard     |  | U+E7B3 | pictWoundHardUp          |
| 'wound-soft     |  | U+E7B7 | pictWoundSoftUp          |
| 'metal          |  | U+E7C7 | pictBeaterMetalUp        |
| 'brass-mallets  |  | U+E7D9 | pictBeaterBrassMalletsUp |
| 'triangle       |  | U+E7D5 | pictBeaterTriangleUp     |
| 'triangle-plain |  | U+E7EF | pictBeaterTrianglePlain  |
| 'wire-brushes   |  | U+E7D7 | pictBeaterWireBrushesUp  |
| 'superball      |  | U+E7AE | pictBeaterSuperballUp    |
| 'mallet         |  | U+E7DF | pictBeaterMallet         |
| 'metal-hammer   |  | U+E7E0 | pictBeaterMetalHammer    |
| 'hammer         |  | U+E7E1 | pictBeaterHammer         |
| 'hand           |  | U+E7E3 | pictBeaterHand           |
| 'finger         |  | U+E7E4 | pictBeaterFinger         |
| 'fist           |  | U+E7E5 | pictBeaterFist           |
| 'fingernails    |  | U+E7E6 | pictBeaterFingernails    |

Electronic Music Symbols

```
\ekm-fader LEVEL ORIENTATION
\ekm-midi LEVEL ORIENTATION
```

Draw a fader (volume control) and a MIDI controller, respectively, as markup.  
LEVEL is drawn as a label next to the control according to ORIENTATION . #f draws no label.

- $\geq 0$  is a percent value.
- $< 0$  is a decibel (dB) value, e.g. -6.0 is equivalent to 50.

For the thumb position, LEVEL is rounded to the nearest integral percent value, limited to 100. If a symbol is defined exactly for this value, this symbol is drawn. Else if an empty control and a thumb symbol are defined, they are combined. Else the symbol for the value nearest to LEVEL is drawn.

Used properties:

- label-format (#f) : #f uses "~a%" for percent and "~adB" for decibel values.
- font-size (0)
- label-size (-4) relative to the font size.
- padding (0.3)

|            |   |        |                      |
|------------|---|--------|----------------------|
| \ekm-fader |   |        |                      |
| 0          |   | U+EB2E | elecVolumeLevel0     |
|            | : |        |                      |
| 100        |   | U+EB33 | elecVolumeLevel100   |
|            |   | U+EB2C | elecVolumeFader      |
|            |   | U+EB2D | elecVolumeFaderThumb |

|           |   |        |                       |
|-----------|---|--------|-----------------------|
| \ekm-midi |   |        |                       |
| 0         |   | U+EB36 | elecMIDIController0   |
|           | : |        |                       |
| 100       |   | U+EB3B | elecMIDIController100 |

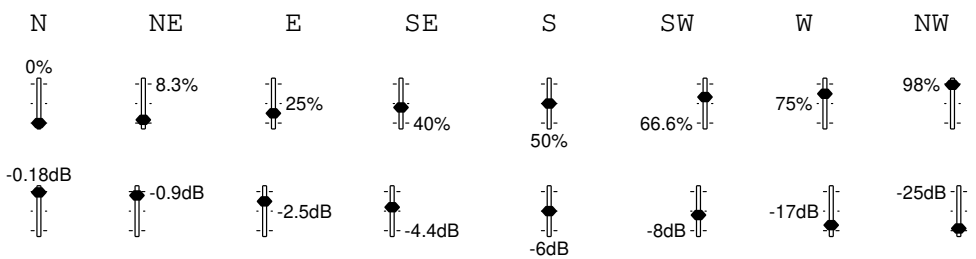
with Ekmelos

|           |  |        |                         |
|-----------|--|--------|-------------------------|
| \ekm-midi |  | U+F6D2 | elecMIDIController      |
|           |  | U+F6D3 | elecMIDIControllerThumb |

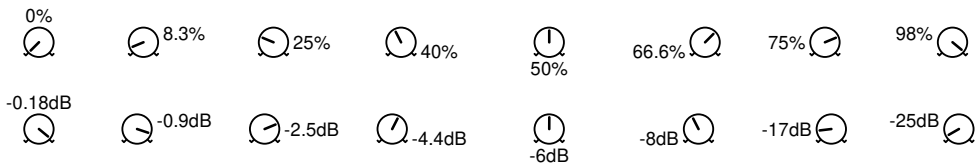
Examples

ORIENTATION

\ekm-fader



\ekm-midi



# Arrows and Arrow Heads

---

`\ekm-arrow` STYLE ORIENTATION

Draw an arrow or arrow head of STYLE according to ORIENTATION as markup.  
Arrows can have glyphs for several orientations. The remaining orientations are achieved by flipping or rotating through 90° or 45°. All following six styles have glyphs for the 8 orientations N ... NW.

## STYLE

|             |   |        |                  |
|-------------|---|--------|------------------|
| 'black      | ↑ | U+EB60 | arrowBlackUp     |
| 'white      | ↑ | U+EB68 | arrowWhiteUp     |
| 'open       | ↑ | U+EB70 | arrowOpenUp      |
| 'black-head | ▲ | U+EB78 | arrowheadBlackUp |
| 'white-head | △ | U+EB80 | arrowheadWhiteUp |
| 'open-head  | ^ | U+EB88 | arrowheadOpenUp  |

`\ekm-arrow-head` AXIS DIRECTION FILLED

Draw an arrow head as markup, i.e. black-head if FILLED is a true value, else open-head .

## with Ekmelos

|                       |   |        |
|-----------------------|---|--------|
| 'simple               | ↑ | U+2191 |
| 'double               | ⇑ | U+21D1 |
| 'triple               | ⇓ | U+290A |
| 'quadruple            | ⇓ | U+27F0 |
| 'black-wide           | ⇑ | U+2B06 |
| 'white-wide           | ⇑ | U+21E7 |
| 'triangle             | ↑ | U+2B61 |
| 'triangle-bar         | ↑ | U+2B71 |
| 'two-headed           | ↕ | U+2BED |
| 'dashed               | ↑ | U+21E1 |
| 'triangle-dashed      | ↑ | U+2B6B |
| 'opposite             | ↕ | U+21C5 |
| 'triangle-opposite    | ↕ | U+2B81 |
| 'paired               | ⇑ | U+21C8 |
| 'triangle-paired      | ⇑ | U+2B85 |
| 'bent-tip             | ↗ | U+21B1 |
| 'long-bent-tip        | ↗ | U+2BA3 |
| 'curving              | ↗ | U+2934 |
| 'equilateral-head     | ▲ | U+2B9D |
| 'three-d-head         | ▲ | U+2B99 |
| 'black-triangle       | ▲ | U+25B2 |
| 'white-triangle       | △ | U+25B3 |
| 'black-small-triangle | ▲ | U+25B4 |
| 'white-small-triangle | ▲ | U+25B5 |
| 'half-circle          | ◐ | U+2BCA |
| 'circle-half-black    | ◑ | U+25D3 |
| 'square-half-black    | ◼ | U+2B12 |
| 'diamond-half-black   | ◔ | U+2B18 |
| 'circle-quarters      | ◐ | U+25D4 |

Other Symbols

---

\ekm-fermata STYLE

Draw a fermata as markup.

Used property:

- direction

STYLE

|             |                                                                                     |        |                        |
|-------------|-------------------------------------------------------------------------------------|--------|------------------------|
| 'default    |    | U+E4C0 | fermataAbove           |
|             |                                                                                     | U+E4C1 | fermataBelow           |
| 'short      |    | U+E4C4 | fermataShortAbove      |
|             |                                                                                     | U+E4C5 | fermataShortBelow      |
| 'long       |    | U+E4C6 | fermataLongAbove       |
|             |                                                                                     | U+E4C7 | fermataLongBelow       |
| 'veryshort  |    | U+E4C2 | fermataVeryShortAbove  |
|             |                                                                                     | U+E4C3 | fermataVeryShortBelow  |
| 'verylong   |    | U+E4C8 | fermataVeryLongAbove   |
|             |                                                                                     | U+E4C9 | fermataVeryLongBelow   |
| 'henzeshort |    | U+E4CC | fermataShortHenzeAbove |
|             |                                                                                     | U+E4CD | fermataShortHenzeBelow |
| 'henzelong  |  | U+E4CA | fermataLongHenzeAbove  |
|             |                                                                                     | U+E4CB | fermataLongHenzeBelow  |

with Ekmelos

|             |                                                                                     |        |                        |
|-------------|-------------------------------------------------------------------------------------|--------|------------------------|
| 'extrashort |  | U+F69E | fermataExtraShortAbove |
|             |                                                                                     | U+F69F | fermataExtraShortBelow |
| 'extralong  |  | U+F6A0 | fermataExtraLongAbove  |
|             |                                                                                     | U+F6A1 | fermataExtraLongBelow  |

\ekm-eyeglasses DIRECTION

Draw eyeglasses as markup.

DIRECTION

|           |                                                                                     |        |                |
|-----------|-------------------------------------------------------------------------------------|--------|----------------|
| any value |  | U+EC62 | miscEyeglasses |
|-----------|-------------------------------------------------------------------------------------|--------|----------------|

with Ekmelos

|       |                                                                                     |        |                     |
|-------|-------------------------------------------------------------------------------------|--------|---------------------|
| RIGHT |  | U+F65F | miscEyeglassesRight |
|-------|-------------------------------------------------------------------------------------|--------|---------------------|

`\ekm-metronome COUNT`

Draw COUNT metronome strokes (tickings) as markup.

Used property:

- `word-space`

with Ekmelos

 `U+F614` `noteTick`

`\ekmMetronome MUSIC`

Attach metronome strokes to each note, chord, or rest in MUSIC as a horizontally centered markup above the staff, using `\ekm-metronome`. The number of strokes equals the number of quarter note values of the respective duration (possibly rounded up).

## Examples

```
\relative c'' {
  \ekmMetronome {
    c4
    c2.
    <g c>1
    R1
  }

  \time 6/4
  \ekmMetronome r4
  \ekmMetronome r1*5/4
}
```

